

Social preferences or moral concerns: What drives rejections in the Ultimatum game?*

Pau Juan-Bartroli[†]
Toulouse School of Economics

José Ignacio Rivero-Wildemaue[‡]
Universidad Católica del Uruguay

Abstract

Rejections of positive offers in the Ultimatum Game have been attributed to different motivations. We show that a model combining social preferences and moral concerns provides a unifying explanation for these rejections while accounting for additional evidence. Under the preferences considered, a positive degree of spite is a necessary and sufficient condition for rejecting positive offers. This indicates that social preferences, rather than moral concerns, drive rejection behavior. This does not imply that moral concerns do not matter. We show that rejection thresholds increase with individuals' moral concerns, suggesting that morality acts as an amplifier of social preferences. Using data from [van Leeuwen and Alger \(2024\)](#), we estimate individuals' social preferences and moral concerns using a finite mixture approach. Consistent with previous evidence, we identify two types of individuals who reject positive offers in the Ultimatum Game, but that differ in their Dictator Game behavior.

JEL codes: C78, D64, D81, D91

Keywords: Bargaining, Social preferences, Morality, Ultimatum Game

*We are grateful to Ingela Alger, Ernesto María Gavassa-Pérez, Emin Karagözoğlu, Enrico Mattia-Salonia, Esteban Muñoz-Sobrado and Juan Sebastián Pereyra for their valuable discussions. We thank Pablo Brañas-Garza and Antonio Espín for kindly sharing their data with us. Pau Juan-Bartroli acknowledges funding from the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation programme (grant agreement No 789111 - ERC EvolvingEconomics).

[†]pau.juanbartroli@tse-fr.eu

[‡]joseignacio.rivero@ucu.edu.uy

1 Introduction

In relevant economic interactions, individuals bargain over how to divide resources. Examples include a buyer and seller negotiating the price of an item, an employer and employee bargaining over wages, and team members determining how to allocate responsibilities. The simplest model to represent these interactions is the Ultimatum Game (UG) (Güth et al., 1982). In this game, the *proposer* offers a division of an endowment to the *responder*. If the *responder* accepts this offer, the division is implemented. If the *proposer* rejects this offer, both individuals receive nothing. If both individuals maximize their material payoff, and this is common knowledge, then the unique Subgame Perfect Nash Equilibrium is one where the *responder* accepts any positive offer and the *proposer* offers the lowest possible positive amount. However, the evidence contradicts this prediction, as responders frequently reject positive offers (Camerer, 2003; Güth and Kocher, 2014). As discussed below, several studies propose different explanations of these rejections (Levine, 1998; Fehr and Schmidt, 1999; Xiao and Houser, 2005; Çelen et al., 2017; Karagözoğlu and Keskin, 2018; Aina et al., 2020).

In this paper, we show that combining social preferences with moral concerns can account for these behaviors while also explaining other observed patterns. Our analysis also provides insights into how social preferences interact with moral concerns. Understanding how different motives interact in shaping behavior remains an underexplored area of research (Gavassa-Pérez, 2022). However, recent evidence suggests that prosociality is often driven by a general moral preference to “do the right thing” (Capraro and Rand, 2018), and that deontological motives coexist with outcome-based concerns (Chen and Schonger, 2022). By integrating both dimensions, our framework captures interactions that single-motive models fail to explain.

Importantly, we do not treat social preferences and moral concerns in isolation. While both concerns have substantial empirical support on their own (e.g. Miettinen et al., 2020), evolutionary analysis suggests that it is their *combination* that is favored by natural selection, especially in small-stakes interactions (Alger et al., 2020). This prediction is consistent with recent experimental evidence (van Leeuwen and Alger, 2024; Alger and Rivero-Wildemaube, 2024). To the best of our knowledge, this is the first theoretical paper to examine bargaining behavior under the joint presence of social preferences and moral concerns. We show that this interaction is crucial to explain the evidence.

Before discussing our main results, we describe individuals’ utility function. Individuals exhibit social preferences when they attach some weight to their opponent’s material

payoff, which may vary depending on whether the opponent’s material payoff exceeds or falls below the individual’s own payoff (Becker, 1976; Fehr and Schmidt, 1999).¹ Moral concerns are modeled through universalization reasoning, where individuals act *as if* they assign some weight to the scenario in which others adopt the same strategy as themselves.^{2,3} Under this reasoning, individuals suffer a disutility when selecting a rejection threshold above their own offer. This occurs because in a hypothetical scenario where an individual is matched with someone mimicking their strategy, the resulting payoff is zero—as offers would be rejected.

Our analysis relies on two features that depart from more standard treatments of the UG (see Section 2.1 for more details). First, instead of focusing on computing the game’s Nash equilibria, which we relegate to Appendix B, we consider the strategy that maximizes an individual’s utility given his (potentially incorrect) beliefs. We adopt this approach as the former results in (i) multiplicity of equilibria, (ii) no rejections in equilibrium, and (iii) the need to assume that all individuals hold correct beliefs.⁴ Second, we consider the UG with *role uncertainty*, where individuals select their offers and rejection thresholds without knowing their realized role and are assigned to each role with equal probability.⁵ This assumption is essential for computing the universalization counterfactual—i.e., “what if others acted like me?”—and enables us to directly study the relationship between individuals’ own offers and their rejection thresholds (see Corollary 3).

Results: Our main result characterizes individuals’ optimal strategy in the UG as a function of their degrees of spite and morality. We show that the resulting strategy partitions the parameter space into three regions (Proposition 1). The optimal strategy gives three

¹In our setting, aversion to disadvantageous inequality is equivalent to spite (i.e., attaching a negative weight to others’ material payoff). This term is also referred to as *envy* (Mui, 1995; Levine, 1998).

²This reasoning can be interpreted in several ways: the individual (i) considers what would happen if all other individuals followed his own course of action; (ii) believes that others are generally similar and will therefore make the same choices (similar to the *false consensus effect*, Ross et al., 1977; Dawes, 1989; Butler et al., 2015); or (iii) thinks his action influences the likelihood that others will select it as well (similar to *magical thinking*, Shafir and Tversky, 1992; Daley and Sadowski, 2017).

³We acknowledge that morality cannot be reduced to universalization reasoning. We focus on this formulation due to its prominence in both theoretical and experimental work. Alger and Weibull (2013) and Alger et al. (2020) establish its evolutionary foundations, showing that universalization reasoning can emerge as an adaptive trait that promotes cooperation and group cohesion. For alternative moral concepts in economics, see Harsanyi (1955), Sen (1977), Roth (2007), Roemer (2010), Smith and Wilson (2019), Gavassa-Pérez (2022) and Drouvelis and Gavassa-Pérez (2024).

⁴Departing from equilibrium analysis aligns with the approach in Aina et al. (2020), which considers a *subjective rationality* condition in which individuals maximize their utility given their beliefs and preferences.

⁵This approach is common in studies examining universalization reasoning in asymmetric games (e.g., Alger and Weibull, 2013; Muñoz Sobrado, 2022; Salonia, 2023; van Leeuwen and Alger, 2024; Rivero-Wildemaue, 2025).

main insights. First, the rejections of positive offers in the UG can be rationalized by spite. Specifically, a positive degree of spite is both a necessary and sufficient condition for selecting a positive rejection threshold. In contrast, a positive degree of morality is not even a necessary condition for doing so. This relates to studies showing the spiteful nature of rejections in the UG (Falk et al., 2005; Herrmann et al., 2008; Brañas-Garza et al., 2014).⁶ We argue that the prosocial and spiteful high-punishers documented in Brañas-Garza et al. (2014) can be reconciled by considering varying levels of spite and universalization reasoning (see Section 4). Specifically, prosocial punishers exhibit high levels of universalization reasoning and low but positive levels of spite, whereas spiteful punishers exhibit high levels of spite and moderate levels of universalization reasoning (see Table 2).

Second, for sufficiently small levels of spite, individuals' offers and rejection thresholds increase in their degree of morality. Thus, although positive rejection thresholds are driven by spite, this does not imply that individuals' degree of morality is irrelevant. Universalization reasoning acts as an *amplifier* of individuals' social preferences, which is a novel observation in the literature. This is consistent with Kimbrough and Vostroknutov (2016) and Capraro and Rodriguez-Lara (2025) who document a positive correlation between rejection thresholds and measures of morality. Third, our theory predicts a relationship between individuals' offers and their rejection thresholds. If an individual assigns a sufficiently large weight to universalization reasoning, he will select an offer above his rejection threshold. This relates to the results in Çelen et al. (2017) and Candelo et al. (2019).⁷

Out-of-sample predictions: While our theoretical model predicts the optimal strategy, identifying which individuals select high rejection thresholds remains an open question. This is because individuals' rejection thresholds can result from different combinations of parameters. To address this, we use data from van Leeuwen and Alger (2024) to compute predicted behavior in both the Dictator Game (DG) and UG, at both the individual and aggregate levels. This approach allows us to examine which types of individuals select positive rejection thresholds in the UG, given their DG behavior, and to infer their un-

⁶This literature cast doubt on the explanation that individuals' (prosocial) preferences for fairness motivate the rejection of low offers in the UG (Fehr and Fischbacher, 2003; Gintis et al., 2003). According to this explanation, individuals who make high transfers in the DG are more likely to select high rejection thresholds.

⁷Çelen et al. (2017) document that individuals are less likely to reject offers that are higher than what they would offer as proposers. Candelo et al. (2019) find that offers and (hypothetical) rejection thresholds are positively correlated. These authors interpret these findings as evidence of reciprocity and the *false consensus effect*, respectively. We argue that universalization reasoning also accounts for these observations.

derlying preferences.⁸ In the three-type model, two types of individuals select positive rejection thresholds: those who transfer nothing in the DG and those who transfer a substantial share of the endowment. These types differ in their social preferences but exhibit similar levels of universalization reasoning.⁹ The predicted behavior aligns with the key insights of our model: (i) rejections in the UG are driven by spite, and (ii) universalization reasoning amplifies social preferences.

Related literature: This paper contributes to the theoretical and experimental literature on rejection thresholds in the UG (see [Camerer \(2003\)](#) and [Güth and Kocher \(2014\)](#) for reviews). Previous studies have attributed such rejections to various motivations, including spite ([Levine, 1998](#)), reciprocity ([Çelen et al., 2017](#)), inequity aversion ([Fehr and Schmidt, 1999](#)), fairness ([Karagözoğlu and Keskin, 2018](#)), anger ([Xiao and Houser, 2005](#)) and frustration ([Aina et al., 2020](#)). We contribute to this literature by showing that adding universalization reasoning to a model of social preferences explains several results in a unifying way, including some not accounted for by previous models. More precisely, we explain (i) the rejection of positive offers by subjects who behave either selfishly or prosocially in the DG, (ii) the correlation between rejection thresholds and measures of morality ([Kimbrough and Vostroknutov, 2016](#); [Capraro and Rodriguez-Lara, 2025](#)), (iii) why individuals are more likely to reject offers if those are lower than the offer they would have selected as proposers ([Çelen et al., 2017](#)), and (iv) the positive correlation between offers and (hypothetical) rejection thresholds ([Candelo et al., 2019](#)).

Finally, this paper contributes to the growing literature using universalization reasoning to explain behavior in several settings. This literature builds on [Alger and Weibull \(2013\)](#) and [Alger and Weibull \(2016\)](#) that show the evolutionary stability of universalization reasoning.¹⁰ This reasoning has been used in theoretical studies to rationalize voting ([Alger and Laslier, 2022](#)), paying taxes ([Muñoz Sobrado, 2022](#)), climate change concerns ([Eichner and Pethig, 2021](#)) and behavior in teams ([Sarkisian, 2017](#)). [Alger and Weibull](#)

⁸We use the DG as it abstracts away from strategic considerations isolating individuals' motives more clearly. Additionally, it enables us to determine whether high rejection thresholds are selected by individuals who make large transfers in the DG, as predicted by models where punishment is driven by prosocial motives.

⁹Our out-of-sample predictions are quantitatively lower than those reported in previous studies (e.g., [Brañas-Garza et al., 2014](#)), with the discrepancy being particularly pronounced for rejection thresholds. We interpret these predictions primarily as a qualitative exercise, given the challenges of comparing estimates from experimental settings that differ from our theoretical framework.

¹⁰See [Alger \(2023\)](#) for a review of the evolutionary foundations of universalization reasoning. For its axiomatic foundations, see [Salonia \(2023\)](#). For another model of universalization reasoning, see [Brekke et al. \(2003\)](#). An alternative way of introducing morality is through the *Kantian equilibrium*, which modifies the equilibrium concept ([Roemer, 2010](#); [Roemer, 2015](#)). See [Dizarlar and Karagözoğlu \(2023\)](#) for the study of bargaining interactions using the Kantian equilibrium.

(2017) analyze universalization reasoning and altruism separately across several games. In addition, the mechanism has been successfully tested in laboratory settings (Miettinen et al., 2020; Juan-Bartroli, 2023; van Leeuwen and Alger, 2024; Alger and Rivero-Wildemaue, 2024). Importantly, van Leeuwen and Alger (2024) structurally estimate a model combining universalization reasoning and social preferences and find that this model is better at explaining behavior than one considering only one of these concerns separately. This evidence provides an important justification for studying the predictions of a model combining both social preferences and universalization reasoning.¹¹

Closer to our work, Juan-Bartroli and Karagözoğlu (2024), Rivero-Wildemaue (2023), and Rivero-Wildemaue (2025) study theoretically the effect of universalization reasoning in bargaining interactions.¹² Juan-Bartroli and Karagözoğlu (2024) study how the equilibrium of a Divide-the-Dollar game changes with universalization reasoning and compare it with other preferences. Rivero-Wildemaue (2023) and Rivero-Wildemaue (2025) analyze the equilibria of bilateral trade games that feature different information asymmetries in sequential and simultaneous settings with individuals exhibiting universalization reasoning. Besides considering the UG and by allowing for potentially incorrect beliefs, we differ from these studies by examining the interaction between universalization reasoning and social preferences. We show that this is crucial to understanding individuals' behavior and deriving novel theoretical predictions.

Outline: The remainder of the paper is organized as follows. Section 2 introduces the theoretical framework. Section 3 presents the theoretical results. Section 4 considers out-of-sample results. Section 5 concludes. All the proofs are in Appendix A.

2 Theoretical Framework

2.1 The Ultimatum Game

The UG is composed of two players, the *proposer* and the *responder*. The *proposer* offers a division of a pie $x_1 \in [0, w]$ of known size $w > 0$. If the *responder* accepts the proposed division, it is implemented: the *proposer* earns $w - x_1$, and the *responder* earns x_1 . If the

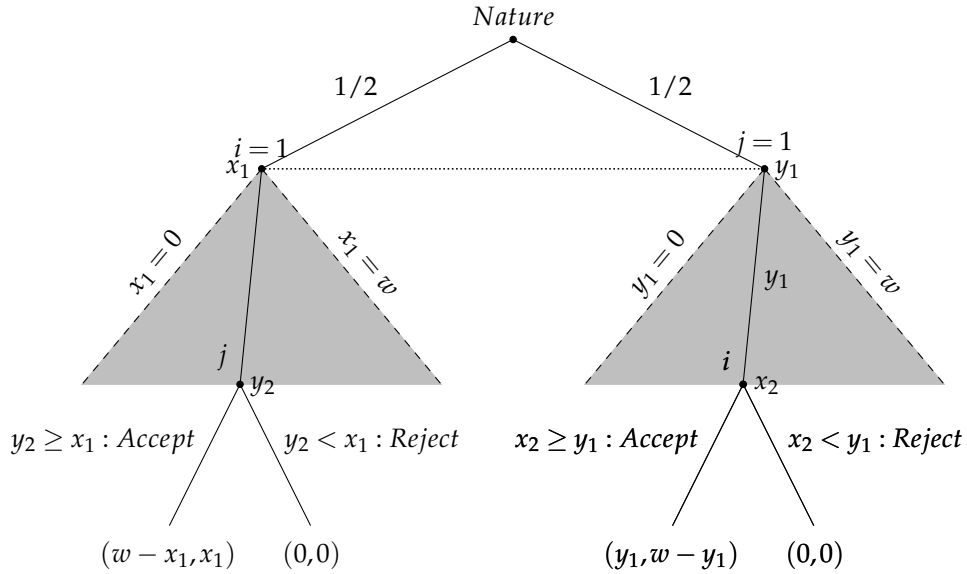
¹¹Alger and Rivero-Wildemaue (2024) show that making role uncertainty salient makes subjects behave more prosocially. They also structurally estimate the intensity of social preferences and universalization reasoning, finding positive and significant values for the latter.

¹²Alger and Weibull (2012) characterize the UG's Nash equilibria when individuals have universalization reasoning. We extend their results by computing the Nash equilibria with individuals who have both universalization and social concerns (see Appendix B).

responder rejects the division, both individuals earn 0. Following previous literature, we assume that responders choose a rejection threshold $x_2 \in [0, w]$ such that any offer below x_2 is rejected, and any offer equal to or above x_2 is accepted.

To introduce universalization reasoning, we consider the ex-ante symmetric version of the UG, where individuals choose their offer and rejection threshold without knowing their role in the interaction. This defines a symmetric interaction in which *Nature* assigns each player a role with equal probability, and then the players learn their respective roles. A strategy is then a tuple $x = (x_1, x_2) \in [0, w]^2$ where x_1 and x_2 represent the offer and the rejection threshold selected by the individual in *proposer* and *responder* roles, respectively. The extensive form of the game is presented in Figure 1.

Figure 1: Ultimatum Game behind the veil of ignorance



Examining the ex-ante symmetric version of the game does not rule out applying our results to settings without role uncertainty. Even a proposer who knows they are not a responder may recognize that roles were randomly assigned and could have been reversed. Consequently, individuals who are certain of their role might still contemplate how choices would differ if roles were reversed and others chose their actions. In this sense, role uncertainty makes universalization reasoning more salient relative to social

preferences (see [Alger and Rivero-Wildemaube, 2024](#)).¹³

Finally, we do not impose that individuals hold correct beliefs about their opponents' strategies. This is motivated by the one-shot, anonymous nature of the interaction, which makes it difficult for individuals to form accurate beliefs about others' actions.¹⁴ In our case, individual i holds beliefs $f_i(y_1)$ and $f_i(y_2)$ on the distributions of offers and rejection thresholds in the population, with $F_i^{y_1}(\cdot)$ and $F_i^{y_2}(\cdot)$ denoting the corresponding cumulative distribution functions. Both $f_i(y_1)$ and $f_i(y_2)$ are assumed to be continuous and smooth functions, which may differ from the true distributions $f(y_1)$ and $f(y_2)$. In Appendix B, we extend [Alger and Weibull \(2012\)](#) to compute the Nash equilibrium of the game when individuals have social preferences and universalization reasoning. The key distinction between computing the strategy that maximizes individuals' utility given their beliefs and deriving the Nash equilibrium of the game is that, in the former, individuals are not required to hold correct beliefs. This assumption allows for rejections of some positive offers as a best response and simplifies the comparative static analysis.

2.2 Preferences

Consider an interaction between two individuals and let X denote the common set of pure strategies, assumed to be non-empty, convex, and compact. Let $\pi : [0, w]^2 \rightarrow \mathbb{R}$ be the individuals' material payoff function, which represents how they evaluate monetary outcomes. Specifically, $\pi(x_i, x_j) \equiv v(\tilde{\pi}(x_i, x_j))$, where $\tilde{\pi}(x_i, x_j)$ is individual i 's monetary payoff under strategy profile (x_i, x_j) , and $v : \mathbb{R} \rightarrow \mathbb{R}$ captures how individuals evaluate monetary payments. We assume that individuals' utility functions combine social preferences and universalization reasoning ([Alger et al., 2020](#); [van Leeuwen and Alger, 2024](#)).

¹³Examining the ex-ante symmetric version of the game admits two interpretations. First, it can be thought of as a mental representation that individuals perform when there is no role uncertainty. For example, the proposer, aware that the roles have been randomly selected, considers how choices would differ if roles were reversed and others chose their actions ([Rawls, 1971](#); [Binmore, 1994](#)). Second, it can be interpreted literally, with choices implemented probabilistically and participants unaware of their roles, as in experiments with role uncertainty ([Andreoni and Bernheim, 2009](#); [Iriberri and Rey-Biel, 2011](#)). Many real-world markets share this feature, where individuals alternate between roles—for instance, sometimes as buyers and sometimes as sellers—from informal exchanges such as flea markets to online marketplaces.

¹⁴It is widely documented that individuals' beliefs are heterogeneous and often inaccurate, even in the simplest experimental settings ([Bellemare et al., 2008](#); [Iriberri and Rey-Biel, 2013](#)).

More concretely, individual i 's utility function is:

$$\begin{aligned}
u_i(x_i, x_j) &= (1 - \kappa_i)\pi(x_i, x_j) \\
&- \alpha_i \max[\pi(x_j, x_i) - \pi(x_i, x_j), 0] \\
&- \beta_i \max[\pi(x_i, x_j) - \pi(x_j, x_i), 0] \\
&+ \kappa_i \pi(x_i, x_i),
\end{aligned} \tag{1}$$

where x_i (resp. x_j) is the strategy of individual i (resp. j), and $\pi(x_i, x_j)$ (resp. $\pi(x_j, x_i)$) is the material payoff of individual i (resp. j) under the strategy profile (x_i, x_j) .

The utility function has three parameters. Two of these parameters capture inequity aversion (as in [Fehr and Schmidt, 1999](#)). The parameter α_i represents i 's (dis)utility from disadvantageous inequality (i.e., obtaining a payoff lower than individual j). Similarly, β_i represents individual i 's (dis)utility from advantageous inequality (i.e., obtaining a payoff higher than individual j). Finally, interpreting the last term in (1) as a representation of Kant's categorical imperative ([Kant, 1785](#)), "act only on the maxim that you would at the same time will to be a universal law", we refer to $\kappa_i \in [0, 1]$ as i 's degree of morality. In this line, $\pi(x_i, x_i)$ measures individual i 's material payoff if individual j used the same strategy as individual i , while κ_i measures the importance he attaches to this counterfactual scenario.¹⁵

Utility function (1) nests several preferences as special cases. For example, setting $\alpha_i = \beta_i = \kappa_i = 0$ represents self-interest. Altruistic preferences ([Becker, 1976](#)) can be achieved by setting $\kappa_i = 0$ and $\alpha_i = -\beta_i$ for a $\beta_i \in (0, 1)$. Inequity aversion preferences ([Fehr and Schmidt, 1999](#)) can be attained by setting $\kappa_i = 0$ and $\alpha_i \geq \beta_i > 0$. Spiteful preferences ([Levine, 1998](#)) can be realized by setting $\kappa_i = 0$ and $\alpha_i = -\beta_i$ for a $\beta_i \in (-1, 0)$. Lastly, *homo moralis* preferences ([Alger and Weibull, 2013](#)) can be obtained by setting $\alpha_i = \beta_i = 0$ and $\kappa_i \in (0, 1]$. Note that (1) does not directly consider reciprocity motives ([Rabin, 1993](#); [Dufwenberg and Kirchsteiger, 2004](#)). Although reciprocity is relevant in the UG, we are interested in studying the relationship between behavior in the UG and the DG, for which reciprocity is silent.

In this setting, [van Leeuwen and Alger \(2024\)](#) is particularly relevant, as the authors conduct a lab experiment to structurally estimate (1). They find that the estimated parameters at the aggregate level satisfy $\hat{\alpha} > 0$, $\hat{\beta} = 0$ and $\hat{\kappa} > 0$, suggesting that the representa-

¹⁵Alternatively, the parameter κ_i can be interpreted in two ways: (i) as the probability that an individual is paired with someone who behaves like them, similar to the *false consensus effect* ([Ross et al., 1977](#); [Dawes, 1989](#); [Butler et al., 2015](#)); or (ii) as a proxy for *magical thinking*, wherein an individual believes that choosing x makes it more likely that others choose x as well ([Shafir and Tversky, 1992](#); [Daley and Sadowski, 2017](#)).

tive agent dislikes disadvantageous inequality, is indifferent with respect to advantageous inequality, and has a positive degree of morality.¹⁶ This motivates our first assumption.

Assumption 1. *For any i , $\beta_i = 0$, $\alpha_i \geq 0$ and $\kappa_i \in [0, 1]$.*

Thus, we assume that individuals are unaffected by advantageous inequality but may be concerned with disadvantageous inequality and assign some weight to universalization reasoning. Our results also extend when $\beta_i \neq 0$, and $\alpha_i < 0$. However, allowing for these cases complicates the exposition of the results without providing additional insight. Footnote (19) clarifies how the optimal strategy changes when $\beta_i \neq 0$, while footnote (22) addresses the case where $\alpha_i < 0$.

3 Theoretical Results

In this section, we derive the optimal strategy of an individual whose utility function combines social preferences and moral concerns, as in equation (1). Section 3.1 introduces Lemmas 1 and 2, which facilitate this characterization. Section 3.2 characterizes the optimal strategy. Section 3.3 presents several corollaries and discusses how they contribute to rationalizing evidence from the experimental literature.

3.1 Preliminary Analysis

We begin by introducing Assumptions 2 and 3, which facilitate the characterization of the optimal strategy. Assumption 2 restricts individuals' beliefs.

Assumption 2. *For any i , $F_i^{y_2}(\frac{w}{2}) = F_i^{y_1}(\frac{w}{2}) = 1$.*

Assumption 2 implies that all individuals believe others' offers and rejection thresholds do not exceed $\frac{w}{2}$, consistent with previous evidence (Camerer, 2003; Güth and Kocher, 2014). In practice, this assumption restricts individuals' optimal strategies to offers and rejection thresholds not exceeding $\frac{w}{2}$. Therefore, spite (i.e., behindness aversion) influences individuals' utility only in the *responder* role, as proposers always receive a payoff at least equal to that of responders.

Assumption 3 imposes structure on the function individuals use to evaluate monetary payoffs resulting from the interaction.

¹⁶It is worth noting that van Leeuwen and Alger (2024) find $\beta_i \neq 0$ and $\alpha_i < 0$ for some individuals when considering estimates at the individual level. In addition, Alger and Rivero-Wildemaube (2024) find $\beta \neq 0$ at the aggregate level.

Assumption 3. *Function v is increasing, strictly concave and differentiable.*

Assumption 3 implies that individuals exhibit decreasing marginal utility of money and are risk-averse behind the veil of ignorance.¹⁷ Under Assumptions 2 and 3, and normalizing $v(0) = 0$, individual i maximizes the following utility function:

$$\begin{aligned} u_i(x_1, x_2) &= (1 - \kappa_i)v(w - x_1)F_{y_2}^i(x_1) \\ &+ \int_{x_2}^{\frac{w}{2}} [(1 - \kappa_i)v(y_1) - \alpha_i(v(w - y_1) - v(y_1))]dF_i^{y_1} \\ &+ \kappa_i \mathbf{1}_{\{x_1 \geq x_2\}} [v(w - x_1) + v(x_1)]. \end{aligned} \quad (2)$$

Assumption 4. *Function u_i is strictly concave on each region $x_1 < x_2$ and $x_1 > x_2$, continuous on the whole domain and differentiable (except possibly on the diagonal $x_1 = x_2$).*

Assumption 4 is a technical assumption that makes the problem concave and well-defined.¹⁸ Individual i 's utility comprises three terms:¹⁹

- The first term represents individual i 's payoff when offering x_1 as *proposer* given his expected distribution of rejection thresholds. Increasing x_1 decreases individual i 's payoff when his offer is accepted but increases the likelihood of this occurring.
- The second term represents i 's (net) payoff when setting a rejection threshold of x_2 as *responder* given his expected distribution of offers. Increasing x_2 decreases the likelihood of accepting an offer and increases the lowest offer accepted.
- The third term captures individual i 's hypothetical payoff if others adopted the same strategy as himself, as implied by the universalization reasoning. That is, the payoff individual i would receive if matched with someone choosing strategy $x = (x_1, x_2)$. The universalization concern implies that individual i is worse off

¹⁷This assumption is also imposed in [Alger and Weibull \(2013\)](#) when analyzing the DG and UG with individuals who employ universalization reasoning. Our results remain robust when assuming a linear function, $v(x) = x$. However, as explained in footnote (29), this assumption is less convenient when analyzing the DG.

¹⁸Concavity is assured if the belief distributions $F_{y_1}^i$ and $F_{y_2}^i$ are absolutely continuous on $[0, w/2]$ with densities f_1, f_2 that do not increase too fast. Formally, there exist finite constants $b_1, b_2 > 0$ such that $\sup_{x \in [0, w/2]} |f_1'(x)| \leq b_1$ and $\sup_{x \in [0, w/2]} |f_2'(x)| \leq b_2$. The benchmark uniform case $F = \mathcal{U}[0, w/2]$ satisfies these conditions with $b_1 = b_2 = 0$. No specific numerical values of b_1, b_2 are required for the results.

¹⁹When $\beta_i > 0$, (2) would have an additional term equal to $-\beta_i \frac{1}{2} [v(w - x_1) - v(x_1)] F_{y_2}^i(x_1)$ capturing the disutility of being ahead of the other player when acting as the *proposer*. This would make $x_s, \tilde{x}_1(\kappa_i)$, and $\hat{x}(\kappa_i, \alpha_i)$ from Definitions 1, 2, and 4 below to depend positively on β_i , but not affect the qualitative patterns described in the main text.

when (i) rejecting an offer he would propose himself (i.e., $x_1 < x_2$), and (ii) offering a division below the equal split (i.e., $x_1 < \frac{w}{2}$).

To characterize individuals' optimal strategy $x^* = (x_1^*, x_2^*)$, it is useful to define the following four objects.

Definition 1. (*Selfish Offer*): $x_s \equiv \arg \max_{x \in [0, w]} \frac{1}{2} v(w - x) F_{y_2}^i(x_1)$.

Definition 2. (*Constrained optimal offer*): $\tilde{x}_1(\kappa_i) \equiv \arg \max_{x_1 \in [0, w]} \frac{1}{2} (1 - \kappa_i) v(w - x_1) F_{y_2}^i(x_1) + \frac{1}{2} \kappa_i [v(w - x_1) + v(x_1)]$.

Definition 3. (*Constrained optimal rejection threshold*): $\underline{x}_2(\kappa_i, \alpha_i) \equiv \arg \max_{x_2 \in [0, w]} \int_{x_2}^{\frac{w}{2}} [(1 - \kappa_i) v(y_1) - \alpha_i (v(w - y_1) - v(y_1))] dF_i^{y_1}$.

Definition 4. (*Symmetric optimal strategy*): $\hat{x}(\kappa_i, \alpha_i) \equiv \arg \max_{x \in [0, w]} u_i(x, x)$.

The *Selfish offer* is the offer that maximizes individual i 's material payoff, given their beliefs about others' rejection thresholds. This is the offer selected by someone with $\kappa_i = \alpha_i = 0$. The *Constrained optimal offer* and the *Constrained optimal rejection threshold* are the offer and the rejection threshold that maximizes individual i 's utility, conditional on the offer exceeding his rejection threshold.

Finally, the *Symmetric optimal strategy* is the strategy that maximizes individual i 's utility, subject to the constraint that the offer and rejection threshold are equal.

To simplify notation, we omit the parameters of the above-defined objects and refer to them simply as x_s , $\tilde{x}_1$, $\underline{x}_2$, and $\hat{x}$. Lemma 1 characterizes the optimal strategy x^* as a function of x_s , $\tilde{x}_1$, $\underline{x}_2$ and $\hat{x}$, showing that it can be divided into three regions.

Lemma 1. Let $x^* = (x_1^*, x_2^*)$ denote the strategy $(x_1, x_2) \in [0, w]^2$ that maximizes (2). Then,

- If $\tilde{x}_1 \geq \underline{x}_2$, then $x_1^* = \tilde{x}_1$ and $x_2^* = \underline{x}_2$.
- If $\tilde{x}_1 < \underline{x}_2$, then
 - If $u(x_s, \underline{x}_2) \geq u(\hat{x}, \hat{x})$, $x_1^* = x_s$ and $x_2^* = \underline{x}_2$.
 - If $u(x_s, \underline{x}_2) < u(\hat{x}, \hat{x})$, $x_1^* = x_2^* = \hat{x} \in [\tilde{x}_1, \underline{x}_2]$.

Proof. All proofs are in Appendix A. □

Intuitively, the individual computes $\tilde{x}_1$ and $\underline{x}_2$ separately. If $\tilde{x}_1 \geq \underline{x}_2$, then, by Definitions 2 and 3, it is optimal to set $x_1^* = \tilde{x}_1$ and $x_2^* = \underline{x}_2$. When $\tilde{x}_1 < \underline{x}_2$, setting $x_1^* = \tilde{x}_1$ and

$x_2^* = \underline{x}_2$ is not optimal given the disutility the individual suffers when setting $x_1^* < x_2^*$. In that case, the individual selects either (i) $x_1^* = x_2^* = \hat{x}$ or (ii) $x_1^* = x_s$ and $x_2^* = \underline{x}_2$ (with $x_1^* < x_2^*$). Lemma 2 defines two functions $\tilde{\alpha}(\kappa_i)$ and $\tilde{\kappa}(\alpha_i)$, which characterize the relationship between $\tilde{x}_1$ and $\underline{x}_2$, and $u(x_s, \underline{x}_2)$ and $u(\hat{x}, \hat{x})$, respectively.

Lemma 2.

- For any $\kappa_i \in [0, 1)$, if $\alpha_i = \tilde{\alpha}(\kappa_i) \equiv (1 - \kappa_i) \frac{v(\tilde{x}(\kappa_i))}{v(w - \tilde{x}(\kappa_i)) - v(\tilde{x}(\kappa_i))}$, then $\tilde{x}_1 = \underline{x}_2$.
- For any $\alpha_i > 0$, there exists $\tilde{\kappa}(\alpha_i)$, such that, if $\kappa_i = \tilde{\kappa}(\alpha_i)$, then $u(\hat{x}, \hat{x}) = u(x_s, \underline{x}_2)$.

Two key remarks are worth emphasizing. First, there exists a unique threshold $\bar{\alpha} \equiv \frac{v(x_s)}{v(w - x_s) - v(x_s)} > 0$ such that $\tilde{\alpha}(0) = \bar{\alpha}$ and $\tilde{\kappa}(\bar{\alpha}) = 0$. Therefore, in the (α, κ) space, $\tilde{\alpha}(\kappa_i)$ and $\tilde{\kappa}(\alpha_i)$ intersect at $(\bar{\alpha}, 0)$ (see Figure 2). Second, as shown in Proposition 1, $\tilde{\alpha}(\kappa_i)$ is relevant for computing the optimal strategy only when $\alpha > \bar{\alpha}$, whereas $\tilde{\kappa}(\alpha_i)$ is relevant only when $\alpha < \bar{\alpha}$. In these regions, $\tilde{\alpha}(\kappa_i)$ is generally decreasing in κ_i , while $\tilde{\kappa}(\alpha_i)$ is generally increasing in α_i .²⁰

3.2 The optimal strategy

Proposition 1. Let $\bar{\alpha} \equiv \frac{v(x_s)}{v(w - x_s) - v(x_s)} > 0$. Let $x^* = (x_1^*, x_2^*)$ denote the strategy $(x_1, x_2) \in [0, w]^2$ that maximizes (2). Then,

- If $\alpha_i \leq \tilde{\alpha}(\kappa_i)$, $x_1^* = \tilde{x}_1$ and $x_2^* = \underline{x}_2$ (with $x_1^* \geq x_2^*$).
- If either (i) $\alpha_i \in (\tilde{\alpha}(\kappa_i), \bar{\alpha})$ or (ii) $\alpha_i > \bar{\alpha}$ and $\kappa_i > \tilde{\kappa}(\alpha_i)$, $x_1^* = x_2^* = \hat{x}$.
- If $\alpha_i > \bar{\alpha}$ and $\kappa_i \leq \tilde{\kappa}(\alpha_i)$, $x_1^* = x_s$ and $x_2^* = \underline{x}_2$ (with $x_1^* < x_2^*$).

Proposition 1 shows that x^* can be divided into three regions depending on the relationship between x_1^* and x_2^* . Figure 2 illustrates these three regions for the case when $w = 10$, $v(x) = \frac{x^{(1-\rho)}}{1-\rho}$ with $\rho = 0.05$, and beliefs on offers and rejection thresholds follow a beta distribution normalized in $[0, \frac{w}{2}]$ (see Figure 7). In Region 1, the optimal strategy is $x_1^* = \tilde{x}_1$ and $x_2^* = \underline{x}_2$, and occurs when $\alpha_i \leq \tilde{\alpha}(\kappa_i)$. In Region 2, the optimal strategy is $x_1^* = x_2^* = \hat{x}$, and occurs when either (i) $\alpha_i \in (\tilde{\alpha}(\kappa_i), \bar{\alpha})$ or (ii) $\alpha_i > \bar{\alpha}$ and $\kappa_i > \tilde{\kappa}(\alpha_i)$. Finally, in Region 3, the optimal strategy is $x_1^* = x_s$ and $x_2^* = \underline{x}_2$, and occurs when $\alpha_i > \bar{\alpha}$ and $\kappa_i < \tilde{\kappa}(\alpha_i)$.

²⁰Although we do not show these comparative static analysis for every possible specification, we find evidence supporting them in most reasonable functional forms.

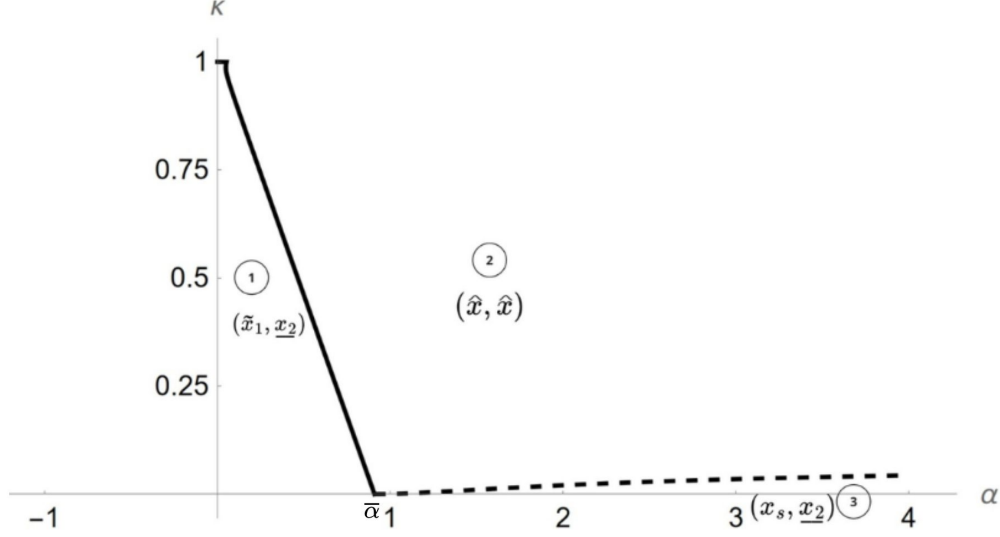


Figure 2: Optimal strategy when $w = 10$, $v(x) = \frac{x^{(1-\rho)}}{1-\rho}$ with $\rho = 0.05$, and $F_{y_2}^i(x) = F_{y_1}^i(x) = F_{[0,w/2]}(2,4)$.

Before proceeding with detailing the strategy of an individual with $\alpha_i > 0$ and $\kappa_i > 0$, it is useful to briefly describe three polar cases:

- **Selfish individual** ($\alpha_i = \kappa_i = 0$): This individual selects $x_1^* = x_s$ and $x_2^* = 0$. This implies that the individual never rejects a positive offer and selects the offer that maximizes his expected payoff given his beliefs about others' rejection behavior.
- **Spiteful individual** ($\alpha_i > 0$ and $\kappa_i = 0$): This individual selects $x_1^* = x_s$ and $x_2^* = \underline{x}_2 > 0$. This implies that the individual selects a strictly positive rejection threshold, which increases with α_i , and the payoff-maximizing offer for any value of α_i .
- **Homo moralis individual** ($\alpha_i = 0$ and $\kappa_i \in (0,1)$): This individual selects $x_1^* = \tilde{x}(\kappa_i)$ and $x_2^* = 0$. This individual never rejects a positive offer, while his offer increases with κ_i .

When the individual combines both $\alpha_i > 0$ and $\kappa_i \in (0,1]$, his strategy is as follows. If the individual is not sufficiently spiteful, i.e., $\alpha_i \leq \bar{\alpha}$, he faces a trade-off between (i) selecting the strategy with the offer and rejection threshold that maximizes constrained utility (i.e., $\tilde{x}_1$ and $\underline{x}_2$) and (ii) adopting a symmetric strategy that accepts offers yielding a negative payoff (i.e., $x_2 < \underline{x}_2$). The resolution of this trade-off depends on α_i . When $\alpha_i < \tilde{\alpha}(\kappa_i)$, then

$\tilde{x}_1 > \underline{x}_2$, which implies that setting $x_1 = \tilde{x}_1$ and $x_2 = \underline{x}_2$ is optimal (see Definitions 2 and 3). On the other hand, when $\alpha_i \in (\tilde{\alpha}(\kappa_i), \bar{\alpha})$, then $\tilde{x}_1 < \underline{x}_2$, which implies that setting $x_1 = \tilde{x}_1$ and $x_2 = \underline{x}_2$ is not optimal, as individual i suffers a disutility when rejecting offers above what she would have offered. In that case, individual i prefers adopting a symmetric strategy with $x_1^* = x_2^* = \hat{x} \in (\tilde{x}_1, \underline{x}_2)$ where the additional disutility is absent.

When the individual is sufficiently spiteful (i.e., $\alpha_i > \bar{\alpha}$), he faces a trade-off between (i) adopting a symmetric strategy with $x_2 < \underline{x}_2$ and (ii) selecting a rejection threshold exceeding their offer in the *proposer* role (i.e., $x_1 < x_2$). The resolution of this trade-off depends on κ_i . To see this, note that all strategies with $x_1 < x_2$ give a payoff of zero if the strategy is universalized (i.e., the third term in (2) is zero). When κ_i is low, the individual selects the strategy that maximizes his (net) payoff (i.e., $x_1^* = x_s$ and $x_2^* = \underline{x}_2$) as the disutility of setting $x_1 < x_2$ is low.²¹ Conversely, when κ_i is high, the individual chooses an offer equal to their rejection threshold (i.e., $x_1^* = x_2^*$), accepting offers that yield a negative (net) payoff (i.e., $x_1^* = x_2^* \in [\tilde{x}_1, \underline{x}_2]$).

Although predictions in all three regions described in Proposition 1 are theoretically possible, the area corresponding to Region 3 appears quantitatively negligible under most reasonable specifications. Moreover, for reasonable values of α_i and κ_i , most individuals fall within Region 1. To illustrate this, we use the individual estimates of α_i and κ_i from van Leeuwen and Alger (2024) to classify subjects into the three regions of Figure 2. Figure 14 (in Appendix C) illustrates that 95% of subjects (91 out of 96) are situated in Region 1 while 5% (5 out of 96) are in Region 2. This classification is quantitatively similar when using other functional forms of utility and beliefs.

3.3 Relation with experimental results

In this section, we relate the optimal strategy characterized in Proposition 1 to findings from the experimental literature. A key question is what defines individuals who reject positive offers. Corollary 1 shows that punishments observed in the UG have a spiteful nature.

Corollary 1. *For any $\kappa_i \in [0, 1)$, $\alpha_i \leq 0$ is a necessary and sufficient condition for $x_2^* = 0$.*

Therefore, a necessary and sufficient condition for rejecting positive offers is having

²¹Conditional on selecting a strategy with $x_1 < x_2$, the individual prefers to set $x_1 = x_s$ as this maximizes his material payoff when proposer.

$\alpha_i > 0$.²² On the other hand, having $\kappa_i > 0$ is not a necessary condition for selecting a rejection threshold above zero. Intuitively, κ_i acts as an amplifier of α_i and is therefore relevant only when $\alpha_i > 0$. Corollary 1 relates to the findings documented in Brañas-Garza et al. (2014), who identify two types of high-punishers: one prosocial type, characterized by high offers in the DG, and one spiteful type, characterized by zero transfers. These results are consistent with two types of individuals: one with high κ_i and low (but positive) α_i , and one with low κ_i and high α_i . Although both types select high rejection thresholds in the UG, the former transfers large amounts in the DG, whereas the latter transfers zero. We confirm this argument in Section 4, where we conduct an out-of-sample exercise with data from van Leeuwen and Alger (2024).

Although $\tilde{x}_1$, $\hat{x}$, and $\underline{x}_2$ are continuous and increasing in κ_i , this is not always the case for x_1^* and x_2^* . Intuitively, x_1^* and x_2^* increase with κ_i when they fall within one of the three regions defined in Proposition 1. However, x_1^* (resp. x_2^*) jumps up (resp. down) when changing from Region 3 to Region 2 (see Figure 3). Corollary 2 shows that having $\alpha_i \leq \bar{\alpha}$ is sufficient to have both x_1^* and x_2^* continuous and increasing in κ_i .

Corollary 2. *When $\alpha_i \in [0, \bar{\alpha})$, x_1^* and x_2^* are continuous in κ_i with $\frac{\partial x_1^*}{\partial \kappa_i} \geq 0$ and $\frac{\partial x_2^*}{\partial \kappa_i} \geq 0$.*

As mentioned before, $\bar{\alpha}$ appears quantitatively large under most reasonable specifications, suggesting that this condition is met by the majority of individuals. Therefore, Corollary 2 aligns with experimental studies that show a positive correlation between rejection thresholds and measures of morality. Kimbrough and Vostroknutov (2016) document a positive correlation between individuals' rejection thresholds and their degree of norm-following.²³ Similarly, Capraro and Rodriguez-Lara (2025) document a positive correlation between individuals' rejection thresholds and two measures of morality (based on a "trade-off" game and a "moral foundations" questionnaire).²⁴

Figure 3 illustrates how the optimal offer and rejection threshold vary with κ_i when (i) $\alpha_i = 0$ (dashed blue), (ii) $\alpha_i = 0.5$ (dotted black), and (iii) $\alpha_i = 3$ (dashed red). These scenarios represent all possible cases: (i) the optimal strategy is always in Region 1 ($\alpha_i =$

²²Note that $x_2^* = 0$ for any $\alpha_i \leq 0$. Thus, the optimal strategy of an individual with $\alpha_i < 0$ coincides with that of an individual with $\alpha_i = 0$. However, when $\kappa_i = 1$ and $\alpha_i = 0$, the individual disregards material payoff and focuses solely on the implications of universalizing their strategy. In this case, $x_1^* = \frac{w}{2}$ and $x_2^* \in [0, \frac{w}{2}]$, implying that x_2^* may be strictly positive.

²³Juan-Bartroli (2023) show that models of norm compliance can be reformulated as models of *homo moralis* preferences, implying a positive correlation between an individual's degree of norm compliance (γ_i) and their degree of morality (κ_i). Furthermore, Juan-Bartroli (2023) documents this positive correlation.

²⁴This complementarity between morality and social preferences relates to the research in psychology showing that individuals with stronger moral identities are more likely to engage in costly prosocial behavior (e.g., Aquino and Reed, 2002).

0), (ii) it moves from Region 1 to Region 2 ($\alpha_i = 0.5$), and (iii) it moves from Region 3 to Region 2 ($\alpha_i = 3$).

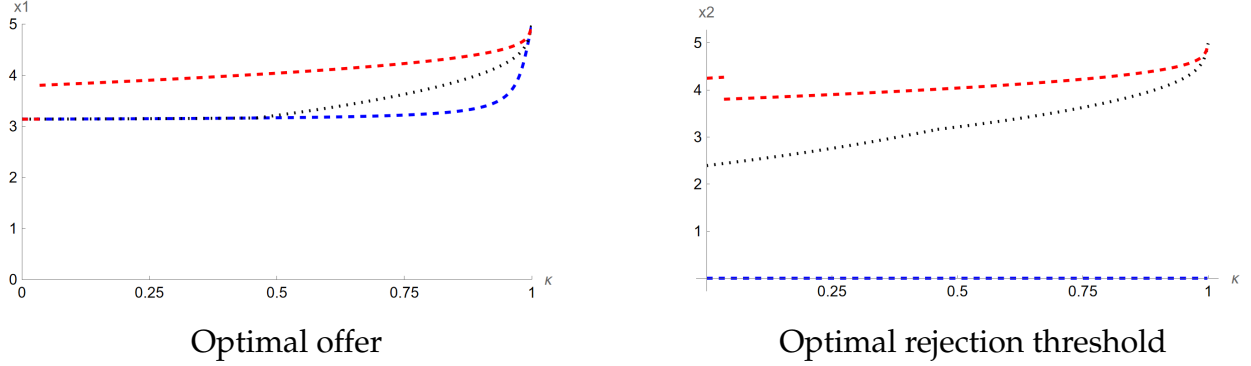


Figure 3: Optimal offer (left) and optimal rejection threshold (right) when $w = 10$, $v(x) = \frac{x^{(1-\rho)}}{1-\rho}$ with $\rho = 0.05$, and $F_{y_2}^i(x) = F_{y_1}^i(x) = F_{[0,w/2]}(2,4)$. $\alpha_i = 0$ (dashed blue), $\alpha_i = 0.5$ (dotted black), and $\alpha_i = 3$ (dashed red).

When $\alpha_i = 0$, $x_1^* = \tilde{x}_1$ and $x_2^* = 0$. Therefore, while x_1^* is increasing in κ_i , x_2^* is equal to zero for any $\kappa_i \in [0, 1]$. When $\alpha_i = 0.5$, the optimal strategy is characterized by two regions divided by $\kappa_i = 0.46$ (i.e., the value such that $\tilde{x}_1 = \underline{x}_2$). When $\kappa_i \in [0, 0.46]$, then $x_1^* = \tilde{x}_1$ and $x_2^* = \underline{x}_2$, both increasing in κ_i . On the other hand, when $\kappa_i \in [0.46, 1]$, then $x_1^* = x_2^* = \hat{x}$, increasing in κ_i . Finally, when $\alpha_i = 3$, the optimal strategy is also characterized by two regions. When $\kappa_i \in [0, 0.03]$, then $x_1^* = x_s$ (constant in κ_i) and $x_2^* = \underline{x}_2$ (increasing in κ_i). When $\kappa_i \in [0.03, 1]$, then $x_1^* = x_2^* = \hat{x}$ (increasing in κ_i). Note that in $\kappa_i = 0.03$ (i.e., the value in which $u(x_s, \underline{x}_2) = u(\hat{x}, \hat{x})$), x_1^* jumps up from x_s to $\hat{x}$, while x_2^* jumps down from $\underline{x}_2$ to $\hat{x}$.

Several studies have examined the relationship between x_1 and x_2 . Corollary 3 shows that if individuals attach a sufficiently large weight to universalization, then their offer is above their rejection threshold.

Corollary 3. *There exists $\tilde{\kappa}(\alpha_i) \geq 0$ such that for any $\kappa_i > \tilde{\kappa}(\alpha_i)$, $x_1^* \geq x_2^*$.*

Corollary 3 can be interpreted as follows. When individuals attach a sufficiently high weight to universalization reasoning, their strategy does not have their offer below their rejection threshold. In other words, individuals avoid proposing offers they would reject in the *responder* role. This finding aligns with Çelen et al. (2017), who document that responders are more likely to reject offers lower than those they would have proposed in the *proposer* role. Similarly, Candelo et al. (2019) finds a positive correlation between offers and hypothetical rejection thresholds, with the latter elicited in a post-experimental

survey. [Çelen et al. \(2017\)](#) and [Candelo et al. \(2019\)](#) interpret these findings as evidence of reciprocity and the *false consensus effect*, respectively. We argue that both observations can be explained through universalization reasoning.

4 Types of punishers

In this section, we use data from [van Leeuwen and Alger \(2024\)](#) to better understand which individuals select high rejection thresholds in the UG. Although our theoretical model predicts individuals’ strategies, high rejection thresholds may arise from different combinations of parameters. In Section 4.1, we use estimates from [van Leeuwen and Alger \(2024\)](#) to compute predicted transfers in the DG and rejection thresholds in the UG at the individual level.²⁵ In Section 4.2, we estimate a finite mixture model to characterize the behavior of different types. In our preferred specification, we identify two distinct types of individuals who select positive rejection thresholds: those who transfer approximately half of the endowment in the DG and those who transfer nothing.

We emphasize that we consider this exercise as qualitative rather than quantitative. This is motivated by two primary considerations. First, our theoretical model and the choice data from [van Leeuwen and Alger \(2024\)](#) differ in several dimensions (discussed below). This requires us to adopt further assumptions to make them directly comparable. Second, prior research highlights the challenges of making accurate quantitative out-of-sample predictions in contexts that differ from those used to estimate individual parameters ([Erev and Roth, 1998](#)).

4.1 Predicted transfers and rejection thresholds

Before proceeding, a brief explanation of [van Leeuwen and Alger \(2024\)](#) is in order. The experiment was conducted at the CentERlab of Tilburg University with 136 subjects, primarily students.²⁶ The experiment elicited subjects’ behavior and beliefs across six sequential prisoner’s dilemmas, six trust games, and six mini-ultimatum games. In each game, subjects made decisions for both roles behind the veil of ignorance. Additionally, subjects reported their beliefs about others’ choices in each game. Each game involved

²⁵Besides the alternative sample restriction detailed in footnote (28), the main difference from the estimation conducted in [van Leeuwen and Alger \(2024\)](#) is that we perform out-of-sample predictions for dictator and ultimatum games.

²⁶The data from [van Leeuwen and Alger \(2024\)](#) used in this paper is available [here](#).

two possible actions, with monetary payoffs varying according to the combination of players' choices.

In the mini-ultimatum games, the proposer chooses between an egalitarian distribution (50,50) and an unequal split favoring themselves. If the proposer selects the unequal split, the responder decides whether to punish the proposer by opting for the egalitarian outcome (10,10) or to accept the proposed unequal distribution. Across six versions of the game, the proposer's unequal offers yielded (60,40), (65,35), (70,30), (75,25), (80,20), and (85,15), respectively. This design allows to compute each subject's *implicit rejection threshold*: the highest payoff in an unequal split that the subject still prefers to reject in favor of (10,10) (more details below).

The authors use this data to estimate equation (1) at various aggregation levels. For each level, they provide distinct sets of estimates based on differing assumptions regarding subjects' risk preferences.²⁷ They estimate their parameters at the experimental currency level (1 point = 0.17€). Accordingly, we set $w = 58.8$ to represent an endowment of 10€.

Given Assumption 3 imposes risk aversion, we focus on the subjects from [van Leeuwen and Alger \(2024\)](#)'s *core sample* whose individual CRRA utility function estimates satisfy $\alpha_i, \beta_i \in [-2, 2]$ and $\kappa_i \in [0, 1]$.²⁸ This results in a sample of 96 subjects, referred to as our *test sample*.

Table 1 presents descriptive statistics for the individual estimates in our test sample (see Figures 8 to 10 for their distributions). On average, subjects in this sample are (i) spiteful when behind (mean α_i is 0.15), (ii) altruistic when ahead (mean β_i is -0.16), and attach a positive weight towards universalization reasoning (mean κ_i is 0.25). Substantial heterogeneity is observed in the estimated individual preference parameters.

Variable	Min	Max	Mean	Median
α	-0.87	1.07	0.15	0.15
β	-1.97	1.12	-0.16	-0.05
κ	0.00	0.89	0.25	0.19

Table 1: Descriptive statistics of individual estimates in the test sample ($N = 96$).

²⁷Their individual-level estimates assume either risk-neutral or CRRA preferences, while their aggregate-level models assume either risk-neutrality or logarithmic utility.

²⁸[van Leeuwen and Alger \(2024\)](#) define their *core sample* as subjects whose individual risk-neutrality utility estimates satisfy $\alpha_i, \beta_i, \kappa_i \in [-2, 2]$, resulting in a sample of 112 subjects. We introduce the additional restriction $\kappa_i \in [0, 1]$ and apply the same criteria under risk aversion.

To compute individuals' predicted transfer and rejection threshold, we assume that they exhibit the preferences described in expression (1), with $v(x) = \ln(x + 1)$.²⁹ The optimal transfer of individual i in the DG is given by:

$$x^* = \arg \max_{x \in [0, w]} \frac{1}{2}(1 - \kappa_i) \cdot v(w - x) - \frac{1}{2}\alpha_i \max\{v(x) - v(w - x), 0\} - \frac{1}{2}\beta_i \max\{v(w - x) - v(x), 0\} + \frac{1}{2}\kappa_i[v(w - x) + v(x)]. \quad (3)$$

Figure 4 shows the distribution of predicted transfers in the DG (see Table 3 for the descriptive statistics). The average predicted transfer is approximately 16 % of the endowment, lower than the 28.35% in Engel (2011) and the 40% in Brañas-Garza et al. (2014). Although the predicted transfers are lower than those reported by Engel (2011), the distribution is qualitatively similar (see Figure 12).

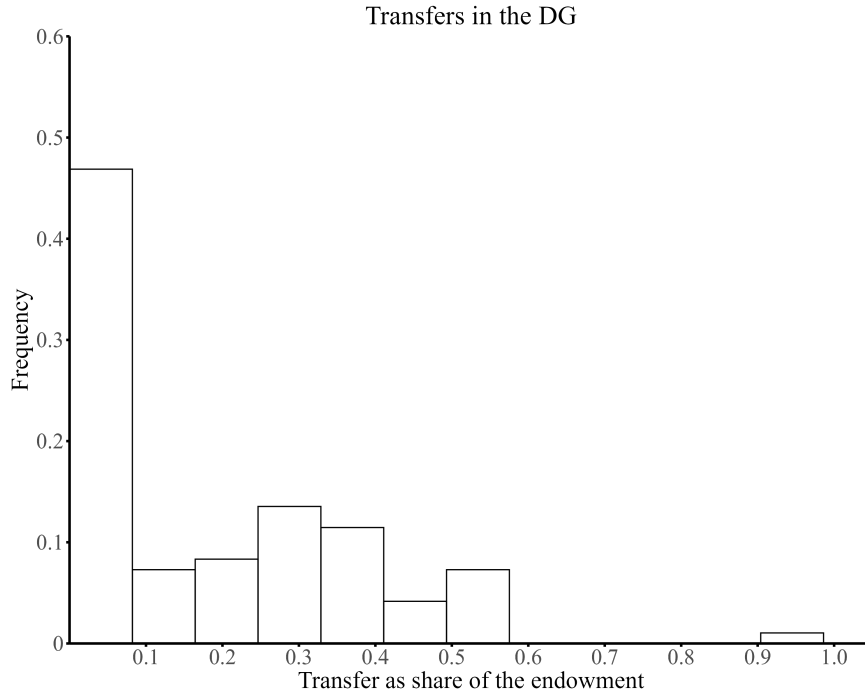


Figure 4: Predicted share transferred in the Dictator Game. Using van Leeuwen and Alger (2024) data, CRRA estimates, for $\alpha, \beta \in [-2, 2]$ and $\kappa \in [0, 1]$. $N = 96$.

²⁹Alternative specifications of $v(x)$ are problematic for several reasons. First, using $v(x) = x$ is not convenient, as transfers in the dictator game would be restricted to either 0 or $\frac{w}{2}$. This prevents us from analyzing rejection thresholds for individuals who transfer intermediate amounts. Second, $v(x) = \frac{x^{1-\rho}}{1-\rho}$ is also problematic, as it implies $v(0) < 0$. Although van Leeuwen and Alger (2024) use this specification, their games involve strictly positive monetary payoffs.

To compute individuals' strategies in the UG, we assume that all individuals select $x_1^* = \frac{w}{2}$. Therefore, individuals' rejection thresholds are given by $x_2^* = \underline{x}_2$, as described in Definition 3. This assumption is justified for three main reasons. First, there is a large concentration of offers in the UG around $\frac{w}{2}$.³⁰ Importantly, $x_1^* = \frac{w}{2}$ may be less influenced by individual preferences, as it can also reflect risk preferences, beliefs about others' rejection thresholds, or one's own hypothetical rejection threshold (see Candelo et al., 2019). Second, setting $x_2^* = \underline{x}_2$ is convenient as $\underline{x}_2$ does not depend on beliefs on others' offers. Third, as shown before, most individuals select a strategy with $x_2^* = \underline{x}_2$ under reasonable specifications (see Figure 14). Figure 5 presents the distribution of predicted rejection thresholds (see Table 3 for the descriptive statistics).

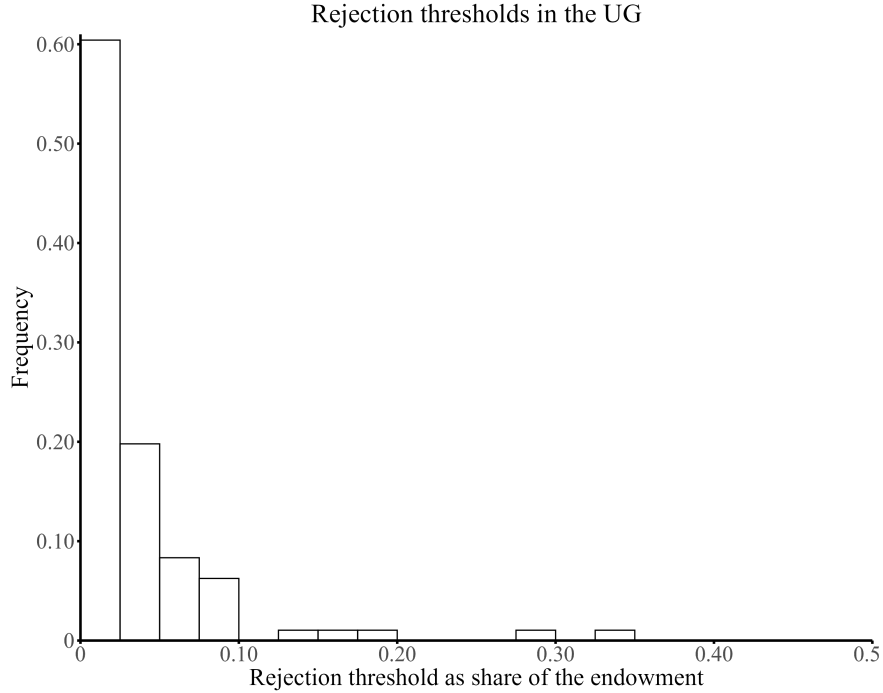


Figure 5: Constrained optimal rejection threshold $\underline{x}_2$ in the UG when $w = 58.8$ (10 euros). Using van Leeuwen and Alger (2024) core sample, CRRA estimates, for $\alpha, \beta \in [-2, 2]$ and $\kappa \in [0, 1]$. $N = 96$.

The average rejection threshold is approximately 3% of the endowment, substantially lower than the 33.35% reported in Brañas-Garza et al. (2014). This low average cannot be attributed to the inclusion of universalization reasoning given that rejection thresholds are increasing in κ_i (see Figure 13).³¹ Despite the low average rejection threshold, more

³⁰For example, Cochard et al. (2021) document a mean offer of 41.5% in a meta-analysis with 96 studies of the UG.

³¹Table 4 shows that a model incorporating social preferences but excluding universalization reasoning predicts an average rejection threshold of 1% of the endowment.

than half of the individuals are willing to reject positive amounts.

We attribute the low rejection thresholds to three main reasons. First, approximately one-third of subjects in our sample exhibit $\alpha < 0$. For these individuals, accepting offers as low as zero is optimal (see Proposition 1 and Figure 13). Second, it is well-documented that people behave more pro-socially when deciding under role uncertainty (Iriberri and Rey-Biel, 2011; Alger and Rivero-Wildemaue, 2024), which is the case in van Leeuwen and Alger (2024). Finally, our estimates are computed from games with binary choices. This may reduce the likelihood of rejections, thereby generating parameter estimates consistent with lower rejection thresholds. Indeed, the *implicit rejection thresholds* observed in van Leeuwen and Alger (2024) are relatively low, with 81% of subjects accepting splits as unequal as 85-15, and an overall acceptance rate of 91% (see Table 1 in van Leeuwen and Alger, 2024). Thus, our predicted behavior is more in line with that observed in van Leeuwen and Alger (2024) than in Brañas-Garza et al. (2014).³²

4.2 3-Type predicted behavior

Previous literature identifies two types of punishers in the UG: *prosocial* punishers, who are willing to sacrifice their own material payoff to enforce fairness norms, and *spiteful* punishers, who assign a negative weight to others' material payoffs. For example, Brañas-Garza et al. (2014) document that individuals who transfer either zero or half of the endowment are most likely to select high rejection thresholds.

In this section, we use the choice data from van Leeuwen and Alger (2024) to structurally estimate the utility parameters (α , β , κ) and the “noise” parameter λ of our test sample ($N = 96$).³³ We use finite mixture models (following Bruhin et al., 2019 and van Leeuwen and Alger, 2024). These models assume that there exists a finite number of types $K \geq 1$ in the population and provide estimates of α , β , κ and λ for each type, along with its population share. We estimate a representative agent model ($K = 1$) and the cases with $K = 2$ and $K = 3$ using the expectation-maximization algorithm (McLachlan et al., 2019).³⁴

³²Figure 15 illustrates the relationship between predicted transfers in the DG and rejection thresholds in the UG. For comparison, actual transfers and thresholds reported in Brañas-Garza et al. (2014) are presented in Figures 20 and 21. Figures 16 to 18 present predicted transfers and rejection thresholds assuming $\kappa = 0$. While predicted transfers in the DG are similar, rejection thresholds display substantially less variability and are more concentrated near zero. Finally, Figure 19 highlights the crucial role of social preferences in explaining differences in rejection thresholds in the UG.

³³This noise parameter λ reflects the probability that decisions are made according to the postulated utility function, with a lower λ indicating that choices follow closely the posited function.

³⁴For a detailed discussion of the method, see van Leeuwen and Alger (2024).

The upper part of Table 2 reports the model results across different levels of aggregation. Our estimates are remarkably similar to those reported in van Leeuwen and Alger (2024) when imposing logarithmic utility (see their Table 5). When assuming a representative agent ($K = 1$), the estimates for the parameters are $\alpha_0 = 0.14$, $\beta_0 = -0.01$, and $\kappa_0 = 0.22$, with the subscript 0 indicating the representative agent. These estimates are consistent with Assumption 1.

When we allow for multiple types, all types exhibit positive degrees of morality ($\kappa > 0$), emphasizing the importance of this concern for understanding individuals' behavior. In the two-type model, Type 1 (61% of the sample) exhibits weak spite both when behind and ahead ($\alpha_1 = 0.05$, $\beta_1 = 0.08$) and strong universalization concerns ($\kappa_1 = 0.25$). Type 2 displays strong spite both when behind and ahead ($\alpha_2 = 0.28$, $\beta_2 = 0.28$) and lower universalization concerns ($\kappa_2 = 0.16$). In the three-type model, Type 1 (29% of the sample) exhibits spite both when behind and ahead ($\alpha_1 = 0.13$, $\beta_1 = 0.22$) and strong universalization concerns ($\kappa_1 = 0.26$). Type 2 (33% of the sample) exhibits no social preferences, with both coefficients not significantly different from zero, and moderate universalization concerns ($\kappa_2 = 0.22$). Finally, Type 3 is essentially identical to Type 2 in the two-type model and accounts for a similarly sized fraction of subjects (38% versus 39% in the two-type model).

Based on the aggregate preference estimates, we compute the predicted transfers in the DG and the rejection thresholds in the UG. The last two rows of Table 2 display these predictions for each type. Given that the three-type model is the preferred specification in van Leeuwen and Alger (2024), we focus our analysis on its results. Moreover, when comparing model fit using the *Integrated Completed Likelihood* (ICL) and the *Normalized Entropy Criterion* (NEC), our findings align with those of van Leeuwen and Alger (2024): the NEC favors the two-type model, while the ICL favors the three-type model.³⁵

³⁵For both metrics, lower values indicate a better model fit. See Celeux and Soromenho (1996) and Biernacki et al. (2000) for detailed derivations of the NEC and ICL metrics, respectively, and McLachlan et al. (2019) for a discussion on using these criteria to evaluate finite mixture models..

Table 2: Aggregate estimates of preference (α , β , κ) and noise (λ) parameters, type shares and predicted transfers and rejection thresholds for [van Leeuwen and Alger \(2024\)](#) core sample, with individual estimates of $\alpha, \beta \in [-2, 2]$ and $\kappa \in [0, 1]$. $N = 96$.

	One type	Two types		Three types		
	Type 1	Type 1	Type 2	Type 1	Type 2	Type 3
α	0.14 (0.03)	0.05 (0.03)	0.28 (0.04)	0.13 (0.08)	-0.02 (0.06)	0.28 (0.03)
β	-0.01 (0.03)	0.08 (0.03)	-0.30 (0.08)	0.22 (0.06)	-0.08 (0.07)	-0.30 (0.09)
κ	0.22 (0.01)	0.25 (0.02)	0.19 (0.02)	0.26 (0.05)	0.22 (0.04)	0.19 (0.02)
λ	0.25 (0.01)	0.28 (0.02)	0.16 (0.02)	0.21 (0.02)	0.33 (0.05)	0.16 (0.02)
Type share	1.00	0.61 (0.05)	0.39 (0.05)	0.29 (0.07)	0.33 (0.08)	0.38 (0.06)
Model fit						
$\ln(\mathcal{L})$	-2,063.28	-1,902.76		-1,865.90		
EN(τ)	0.00	4.00		13.50		
ICL	4,144.82	3,850.59		3,809.21		
NEC	—	0.02		0.07		
Predicted behavior						
DG transfer	7.80 (0.13)	14.90 (0.25)	0.00 (0.00)	22.10 (0.38)	6.17 (0.10)	0.00 (0.00)
UG rejection threshold	0.84 (0.01)	0.31 (0.01)	1.85 (0.03)	0.85 (0.01)	0.00 (0.00)	1.81 (0.03)

Notes: The table reports parameter estimates for one-, two-, and three-type models. Bootstrapped standard errors (SE) are displayed in parentheses next to the estimates. The DG transfer and UG rejection threshold values are shown as shares of the total endowment in parentheses. $\ln(\mathcal{L})$: Log-likelihood; EN(τ): Entropy; ICL: Integrated completed likelihood; NEC: Normalized entropy criterion

In the three-type model, two types of individuals select positive rejection thresholds: one that transfers nearly 40% of the endowment (Type 1) and another that transfers zero (Type 3). These two types differ primarily in their social preferences. Type 3 exhibits twice the level of spitefulness as Type 1. In contrast, Type 2, characterized solely by universalization reasoning, transfers 10% of the endowment and accepts any offer in the UG. This is in line with the behavior documented in [Brañas-Garza et al. \(2014\)](#).

The predicted behavior reported in Table 2 aligns with the key insights from Section 3: (i) rejections in the UG are driven by spite, and (ii) universalization reasoning amplifies social preferences. Specifically, individuals with small but positive α_i and relatively large κ_i (e.g., Type 1 in the two-type model) exhibit high transfers and positive rejection thresholds. In contrast, individuals with high and positive α_i and β_i but moderate κ_i (e.g., Type 2 in the two-type model or Type 3 in the three-type model) display no transfers in the DG but maintain positive rejection thresholds in the UG.

5 Conclusions

In this paper, we study the motivations behind individuals' rejections in the UG. We characterize the optimal offers and rejection thresholds of individuals whose utility function combine social preferences and moral concerns. We derive three main theoretical results. First, spite plays a crucial role in the rejections observed. Specifically, under the preferences considered, a positive degree of spite is both a necessary and sufficient condition for selecting a positive rejection threshold. In contrast, a positive degree of universalization reasoning is not even necessary for doing so. Second, both offers and rejection thresholds are increasing with individuals' universalization reasoning. Intuitively, universalization reasoning amplifies social preferences and is therefore only active when such preferences are present. Third, when individuals assign a sufficiently large weight to universalization reasoning, their offers exceed their rejection thresholds.

These three predictions are consistent with existing empirical evidence. They help explain (i) the rejection of positive offers by individuals exhibiting either selfish or prosocial behavior in the DG, (ii) the correlation between rejection thresholds and measures of morality (Kimbrough and Vostroknutov, 2016; Capraro and Rodriguez-Lara, 2025), (iii) why individuals are more likely to reject offers if those are lower than the offer they would have selected as proposers (Çelen et al., 2017), and (iv) the positive correlation between offers and (hypothetical) rejection thresholds (Candelo et al., 2019).

To better understand the motives behind individuals' rejections, we use the choice data from van Leeuwen and Alger (2024) to estimate a finite mixture model that characterizes different behavioral types. This approach allows us to predict each type's transfer in the DG and rejection threshold in the UG. Focusing on the three-type model, the data clearly identifies two types who reject offers in the UG and differ in their predicted behavior in the DG. The first type is predicted to transfer nearly 40% of the endowment, while the second type is predicted to transfer nothing. These two types differ primarily in their degree of spite: although both exhibit spiteful concerns, the second type is approximately twice as spiteful as the first. In contrast, the third type, labeled the "non-punisher", lacks social preferences but exhibits universalization reasoning.

Appendices

Appendix A Mathematical proofs

A.1 Proofs main text

When individual i selects strategy $x = (x_1, x_2)$, she gets an expected utility of

$$\begin{aligned} u_i(x_1, x_2) &= (1 - \kappa_i)v(w - x_1)F_{y_2}^i(x_1) \\ &+ \int_{x_2}^{\frac{w}{2}} [(1 - \kappa_i)v(y_1) - \alpha_i(v(w - y_1) - v(y_1))] dF_i^{y_1} \\ &+ \kappa_i \mathbf{1}_{\{x_1 \geq x_2\}} [v(w - x_1) + v(x_1)], \end{aligned} \quad (4)$$

In the main text, we defined the four following objects.

- **Definition 1** (*Selfish Offer*): $x_s \equiv \arg \max_{x \in [0, w]} \frac{1}{2}v(w - x_1)F_{y_2}^i(x_1)$.
- **Definition 2** (*Constrained optimal offer*): $\tilde{x}_1(\kappa_i) \equiv \arg \max_{x_1 \in [0, w]} \frac{1}{2}(1 - \kappa_i)v(w - x_1)F_{y_2}^i(x_1) + \frac{1}{2}\kappa_i(v(w - x_1) + v(x_1))$.
- **Definition 3** (*Constrained optimal rejection threshold*): $\underline{x}_2(\kappa_i, \alpha_i) \equiv \arg \max_{x_2 \in [0, w]} \int_{x_2}^{\frac{w}{2}} [(1 - \kappa_i)v(y_1) - \alpha_i(v(w - y_1) - v(y_1))] dF_i^{y_1}$.
- **Definition 4** (*Symmetric optimal strategy*): $\hat{x}(\kappa_i, \alpha_i) \equiv \arg \max_{x \in [0, w]} u_i(x, x)$.

As in the main text, we omit the parameters from the above-defined objects and refer to them as x_s , $\tilde{x}_1$, $\underline{x}_2$, and $\hat{x}$.

In Assumption 4, we assumed that $u_i(x_1, x_2)$ is strictly concave in x_1 (i.e., increasing in x_1 over the interval $[0, \tilde{x}_1)$ and decreasing in x_1 over the interval $(\tilde{x}_1, w]$). We now show under which conditions this is the case. To do so, it is sufficient to show that when $x_1 \geq x_2$, $\frac{\partial^2 u_i(x_1, x_2)}{\partial^2 x_1} < 0$.

$$\begin{aligned} \frac{\partial u_i(x_1, x_2)}{\partial x_1} &= -\frac{1}{2}(1 - \kappa_i)v'(w - x_1)F_{y_2}^i(x_1) + \frac{1}{2}(1 - \kappa_i)v(w - x_1)f_{y_2}^i(x_1) \\ &+ \frac{1}{2}\kappa_i[v'(x_1) - v'(w - x_1)], \end{aligned} \quad (5)$$

$$\begin{aligned} \frac{\partial^2 u_i(x_1, x_2)}{\partial^2 x_1} &= \frac{1}{2}(1 - \kappa_i)v''(w - x_1)F_{y_2}^i(x_1) - \frac{1}{2}(1 - \kappa_i)v'(w - x_1)f_{y_2}^i(x_1) \\ &+ \frac{1}{2}(1 - \kappa_i)v(w - x_1)\frac{\partial f_{y_2}^i(x_1)}{\partial x_1} + \frac{1}{2}\kappa_i[v''(w - x_1) + v''(x_1)]. \end{aligned} \quad (6)$$

All terms in (6), except for the third, are strictly negative. Therefore, $\frac{\partial^2 u_i(x_1, x_2)}{\partial^2 x_1} < 0$ as long as $\frac{\partial f_{y_2}^i(x_1)}{\partial x_1}$ is below a certain positive constant, as discussed in footnote 18.

Proof. Lemma 1: Consider two cases: i) $\tilde{x}_1 \geq \underline{x}_2$ and ii) $\tilde{x}_1 < \underline{x}_2$.

For the first case, $x_1^* = \tilde{x}_1$ and $x_2^* = \underline{x}_2$. This follows from the definitions of $\tilde{x}_1$ and $\underline{x}_2$, as conditional on having $\tilde{x}_1 \geq \underline{x}_2$, $\tilde{x}_1$ and $\underline{x}_2$ are the offer and rejection threshold that maximizes individual i 's utility.

For the second case, we first show the three following claims. In all claims, we fix α_i and $\kappa_i \in [0, 1]$ such that $\underline{x}_2 > \tilde{x}_1$.

Claim 1: $x_1^* \leq x_2^*$.

By contradiction, suppose that there exists a strategy x^* with $x_1^* > x_2^*$. By the definition of $\underline{x}_2$, $x_2^* = \underline{x}_2$. Additionally, by the strictly concavity of $u_i(x_1, x_2)$ on x_1 , when $x_1 > x_2$ $u_i(x_1, x_2)$ is strictly decreasing in x_1 for any $x_1 > \tilde{x}_1$. But then, when $x_2^* = \underline{x}_2 > \tilde{x}_1$, choosing $x_1^* = x_2^*$ gives a strictly larger utility than choosing any $x_1^* > x_2^*$.

Claim 2: If $x_1^* < x_2^*$, then $x_1^* = x_s$ and $x_2^* = \underline{x}_2$.

Note that all strategies x that satisfy $x_1 < x_2$ have the same value in the third term in (4) (as $\mathbf{1}_{\{x_1 \geq x_2\}} = 0$). Therefore, conditional on choosing a strategy x with $x_1 < x_2$, the individual selects the offer and rejection threshold that maximizes his expected payoff (i.e., $x_1^* = x_s$ and $x_2^* = \underline{x}_2$).

Claim 3: If $x_1^* = x_2^*$, then $x_1^* = x_2^* \in [\tilde{x}_1, \underline{x}_2]$.

We need to show that there cannot exist a strategy x^* with $x_1^* = x_2^* < \tilde{x}_1$ or $x_1^* = x_2^* > \underline{x}_2$. We rule out the first case by contradiction. But if $x_1^* = x_2^* < \tilde{x}_1$, then $x = (\tilde{x}_1, x_2^*)$ gives a strictly higher utility to the individual as $x_1 = \tilde{x}_1$ maximizes his utility when $x_1 \geq x_2$. We also rule out the second case by contradiction. But if $x_1^* = x_2^* > \underline{x}_2$, then $x = (x_1^*, \underline{x}_2)$ gives a strictly larger utility to the individual as $x_2 = \underline{x}_2$ maximizes his utility when $x_1 \geq x_2$.

With the three previous claims, we can conclude that to compute x^* , when $\underline{x}_2 > \tilde{x}_1$, we need to compare i) $u(x_s, \underline{x}_2)$ and ii) $u(\hat{x}, \hat{x}) = \arg \max_{y \in [\tilde{x}_1, \underline{x}_2]} u(y, y)$. When $u(x_s, \underline{x}_2) > u(\hat{x}, \hat{x})$, then $x_1^* = x_s$ and $x_2^* = \underline{x}_2$, while $x_1^* = x_2^* = \hat{x}$ otherwise. $\square$

Proof. Lemma 2: We first show that there exists $\tilde{\kappa}(\alpha_i)$ such that: (i) when $\kappa_i < \tilde{\kappa}(\alpha_i)$, $u(x_s, \underline{x}_2) > u(\hat{x}, \hat{x})$, (ii) when $\kappa_i = \tilde{\kappa}(\alpha_i)$, $u(x_s, \underline{x}_2) = u(\hat{x}, \hat{x})$, and (iii) when $\kappa_i > \tilde{\kappa}(\alpha_i)$, $u(x_s, \underline{x}_2) < u(\hat{x}, \hat{x})$.

To show the result, three remarks are important to emphasize. First,

$$\begin{aligned} u(x_s, \underline{x}_2) &= \frac{1}{2}(1 - \kappa_i)v(w - x_s)F_{y_2}^i(x_1) \\ &+ \frac{1}{2} \int_{\underline{x}_2}^{\frac{w}{2}} [(1 - \kappa_i + \alpha_i)v(y_1) - \alpha_i v(w - y_1)] f^i(y_1) dy_1 \end{aligned} \quad (7)$$

is strictly decreasing in κ_i . By the Envelope theorem,

$$\frac{\partial u(x_s, \underline{x}_2)}{\partial \kappa_i} = -\frac{1}{2}v(w - x_s)F_{y_2}^i(x_1) - \frac{1}{2} \int_{\underline{x}_2}^{\frac{w}{2}} v(y_1) f^i(y_1) dy_1 < 0 \quad (8)$$

Second, when $\kappa_i = 0$, $u(x_s, \underline{x}_2) \geq u(\hat{x}, \hat{x})$. Third, if $x_s < \frac{w}{2}$, $u(x_s, \underline{x}_2) < u(\hat{x}, \hat{x})$ when $\kappa_i = 1$.³⁶ Then, we need to determine how $u(\hat{x}, \hat{x})$ depends on κ_i . Recall that $u(\hat{x}, \hat{x})$ is defined as follows:

$$\begin{aligned} u(\hat{x}, \hat{x}) &= \frac{1}{2}(1 - \kappa_i)v(w - \hat{x})F_{y_2}^i(\hat{x}) \\ &+ \frac{1}{2} \int_{\hat{x}}^{\frac{w}{2}} [(1 - \kappa_i + \alpha_i)v(y_1) - \alpha_i v(w - y_1)] f^i(y_1) dy_1 \\ &+ \frac{1}{2}\kappa_i [v(w - \hat{x}) + v(\hat{x})], \end{aligned} \quad (9)$$

where $\hat{x} = \arg \max_{y \in [\bar{x}_1, \underline{x}_2]} u(y, y)$. By the Envelope Theorem,

$$\begin{aligned} \frac{\partial u(\hat{x}, \hat{x})}{\partial \kappa_i} &= -\frac{1}{2}v(w - \hat{x})F_{y_2}^i(\hat{x}) \\ &- \frac{1}{2} \int_{\hat{x}}^{\frac{w}{2}} v(y_1) f^i(y_1) dy_1 \\ &+ \frac{1}{2} [v(w - \hat{x}) + v(\hat{x})]. \end{aligned} \quad (10)$$

³⁶When $x_s = \frac{w}{2}$, then $x_1 = \frac{w}{2}$ and $x_2 = \underline{x}_2$ is the optimal strategy for any κ_i and α_i .

To determine the sign of $\frac{\partial u(\hat{x}, \hat{x})}{\partial \kappa_i}$, we examine its derivative:

$$\begin{aligned} \frac{\partial^2 u(\hat{x}, \hat{x})}{\partial^2 \kappa_i} &= \frac{1}{2} \frac{\partial \hat{x}}{\partial \kappa_i} \left[v'(w - \hat{x}) F_{y_2}^i(\hat{x}) - v(w - \hat{x}) f_{y_2}^i(\hat{x}) \right] \\ &+ \frac{1}{2} \frac{\partial \hat{x}}{\partial \kappa_i} v(\hat{x}) f^i(\hat{x}) \\ &+ \frac{1}{2} \frac{\partial \hat{x}}{\partial \kappa_i} [v'(\hat{x}) - v'(w - \hat{x})]. \end{aligned} \quad (11)$$

The sign of $\frac{\partial^2 u(\hat{x}, \hat{x})}{\partial^2 \kappa_i}$ depends on the sign of $\frac{\partial \hat{x}}{\partial \kappa_i}$. We know that $\hat{x}$ is computed from

$$\begin{aligned} \max_{y \in [\tilde{x}_1, \underline{x}_2(\gamma_i, \alpha_i)]} & \frac{1}{2} (1 - \kappa_i) v(w - y) F_{y_2}^i(y) \\ &+ \frac{1}{2} \int_y^{\frac{w}{2}} [(1 - \kappa_i + \alpha_i) v(y_1) - \alpha_i v(w - y_1)] f^i(y_1) dy_1 \\ &+ \frac{1}{2} \kappa_i [v(w - y) + v(y)]. \end{aligned} \quad (12)$$

The first-order condition of the previous expression is given by

$$\begin{aligned} \frac{1}{2} (1 - \kappa_i) [-v'(w - \hat{x}) F_{y_2}^i(\hat{x}) + v(w - \hat{x}) f_{y_2}^i(\hat{x})] \\ - \frac{1}{2} f_{y_1}^i(\hat{x}) [v(\hat{x}) (1 - \kappa_i + \alpha_i) - \alpha_i v(w - \hat{x})] \\ + \frac{1}{2} \kappa_i [v'(\hat{x}) - v'(w - \hat{x})]. \end{aligned} \quad (13)$$

Assumption 4 guarantees that $\hat{x}$ is a maximum of $u(\hat{x}, \hat{x})$. Following previous derivations, we compute the implicit derivative (13) to obtain that $\frac{\partial \hat{x}}{\partial \kappa_i} \geq 0$. This implies that $u(\hat{x}, \hat{x})$ is convex in κ_i . This, jointly with (i) $u(x_s, \underline{x}_2) \geq u(\hat{x}, \hat{x})$ when $\kappa_i = 0$ and (ii) $u(x_s, \underline{x}_2) < u(\hat{x}, \hat{x})$ when $\kappa_i = 1$ guarantees the existence of a unique $\tilde{\kappa}(\alpha_i) \geq 0$ such that when $\kappa_i < \tilde{\kappa}(\alpha_i)$, $u(x_s, \underline{x}_2) > u(\hat{x}, \hat{x})$, and $u(x_s, \underline{x}_2) < u(\hat{x}, \hat{x})$ otherwise. To see this, note that $u(\hat{x}, \hat{x})$ convex implies that there can only exist a unique global minimum of $u(\hat{x}, \hat{x})$ (e.g. at $x_1 = x_2 = \hat{x}(\tilde{\kappa})$ with $\tilde{\kappa} > 0$). Then, $u(\hat{x}, \hat{x})$ is decreasing for $\kappa_i \in [0, \tilde{\kappa})$ and increasing for $\kappa_i > \tilde{\kappa}$. Thus, $u(x_s, \underline{x}_2) \geq u(\hat{x}, \hat{x})$ when $\kappa_i = 0$ implies that $u(x_s, \underline{x}_2) > u(\hat{x}, \hat{x})$ when $\kappa_i = \tilde{\kappa}$, which shows the result.

We now show that there exists a function $\tilde{\alpha}(\kappa_i) \equiv \frac{(1 - \kappa_i)v(\tilde{x}(\kappa_i))}{v(w - \tilde{x}(\kappa_i)) - v(\tilde{x}(\kappa_i))} > 0$ such that: (i) when $\alpha_i = \tilde{\alpha}(\kappa_i)$, $\tilde{x}_1(\kappa_i) = \underline{x}_2$, (ii) when $\alpha_i > \tilde{\alpha}(\kappa_i)$, $\tilde{x}_1(\kappa_i) < \underline{x}_2$, and (iii) when $\alpha_i < \tilde{\alpha}(\kappa_i)$, $\tilde{x}_1(\kappa_i) > \underline{x}_2$. Note that $\tilde{x}_1(\kappa_i) = \underline{x}_2$ is equivalent to $v(\tilde{x}_1)(1 + \alpha_i - \kappa_i) - \alpha_i v(w - \tilde{x}_1) = 0$, or equivalently $\alpha_i = \frac{(1 - \kappa_i)v(\tilde{x}(\kappa_i))}{v(w - \tilde{x}(\kappa_i)) - v(\tilde{x}(\kappa_i))}$. Fixing κ_i such that $\tilde{x}_1(\kappa_i) = \underline{x}_2$, increasing α_i , increases

$\underline{x}_2$, but not $\tilde{x}_1(\kappa_i)$. Thus, for any $\alpha > \tilde{\alpha}(\kappa_i)$, $\tilde{x}_1(\kappa_i) < \underline{x}_2$.³⁷ □

Proof. Proposition 1: Follows from combining Lemmas 1 and 2. □

Proof. Corollary 1: To show the first result, we fix $\alpha_i = 0$. Individual i maximizes

$$\begin{aligned} & \max_{\{x_1, x_2\} \in [0, w]^2} \frac{1}{2}(1 - \kappa_i)v(w - x_1)F_{y_2}^i(x_1) + \frac{1}{2} \int_{x_2}^{\frac{w}{2}} (1 - \kappa_i)v(y_1)f^i(y_1)dy_1 \\ & + \kappa_i \mathbf{1}_{\{x_1 \geq x_2\}} \frac{1}{2} [v(w - x_1) + v(x_1)]. \end{aligned} \quad (14)$$

The first term in (14) does not depend on x_2 , the second term is maximized at $x_2 = 0$, and the third term is maximized at any $x_2 \in [0, x_1]$. Thus, when $\alpha_i = 0$ and $\kappa_i \in [0, 1)$, choosing $x_2 = 0$ maximizes individual's utility. When $\kappa_i = 1$, then $x_1^* = \frac{w}{2}$ and $x_2^* \in [0, \frac{w}{2}]$. Therefore, it is not true that when $\alpha = 0$ and $\kappa = 1$, the optimal strategy cannot have a rejection threshold above 0.

We now show that when $\alpha_i > 0$, $x_2^* > 0$ for any $\kappa_i \in [0, 1)$. By contradiction, suppose that an individual with $\alpha_i > 0$ and $\kappa_i \in [0, 1)$ has an optimal strategy x^* with $x_2^* = 0$. By Definition 3, we know that $\underline{x}_2(\kappa_i, \alpha_i) > 0$ when $\alpha_i > 0$ and $\kappa_i \in [0, 1)$. Therefore, by Proposition 1, x^* must be symmetric (as otherwise $x_2^* > 0$). However, selecting $x_1^* = x_2^* = 0$ is not an optimal strategy as would be dominated by, e.g., $x_1^* = x_s$ and $x_2^* = 0$.

Corollary 2: We distinguish two cases depending if (i) $\alpha_i \in (\tilde{\alpha}(\kappa_i), \bar{\alpha})$ or if (ii) $\alpha_i \in (0, \tilde{\alpha}(\kappa_i))$.

By Proposition 1, if $\alpha_i \in (\tilde{\alpha}(\kappa_i), \bar{\alpha})$, then $x^* = (\hat{x}, \hat{x})$. $\frac{\partial \hat{x}}{\partial \kappa_i} \geq 0$ follows from taking the implicit derivative of (13).

By Proposition 1, if $\alpha_i \in (0, \tilde{\alpha}(\kappa_i))$, $\tilde{x}_1 \geq \underline{x}_2$. We need to show that (i) $\frac{\partial \tilde{x}_1}{\partial \kappa_i} \geq 0$ and (ii) $\frac{\partial \underline{x}_2}{\partial \kappa_i} \geq 0$.

$$\frac{\partial \tilde{x}_1}{\partial \kappa_i} = \frac{-\kappa_i [v'(\tilde{x}_1) - v'(w - \tilde{x}_1)]}{(1 - \kappa_i) \left[v''(w - \tilde{x}_1)F_{y_2}^i(\tilde{x}_1) - v'(w - \tilde{x}_1)f_{y_2}^i(\tilde{x}_1) + v(w - \tilde{x}_1)\frac{\partial f_{y_2}^i(\tilde{x}_1)}{\partial x_1} \right] + \kappa_i [v''(w - \tilde{x}_1) + v''(\tilde{x}_1)]} \geq 0, \quad (15)$$

as both the numerator and denominator are negative.³⁸

$$\frac{\partial \underline{x}_2}{\partial \kappa_i} = \frac{(1 - \kappa_i)\alpha_i [v(w - \underline{x}_2) - v(\underline{x}_2)]}{(1 - \kappa_i)v'(\underline{x}_2) + \alpha_i [v'(\underline{x}_2) + v'(w - \underline{x}_2)]} \geq 0, \quad (16)$$

³⁷Note that $\tilde{\alpha}(\kappa_i) > 0$ as for any $\kappa < 1$, $\tilde{x}_1(\kappa) \in [x_s, \frac{w}{2})$ (and therefore $v(w - \tilde{x}(\kappa)) - v(\tilde{x}(\kappa)) > 0$).

³⁸The numerator is negative as $v'(\tilde{x}_1) - v'(w - \tilde{x}_1) > 0$, while the denominator is negative because of Assumption 4 (see equation (6)).

as both the numerator and denominator are positive.³⁹ $\square$

Proof. Corollary 3: By Proposition 1, $x_2^* > x_1^*$ when $\alpha_i > \bar{\alpha}$ and $\kappa_i \leq \tilde{\kappa}(\alpha_i)$. Therefore, for any $\kappa_i > \tilde{\kappa}(\alpha_i)$, $x_1^* \geq x_2^*$. $\square$

Appendix B Nash Equilibrium in the Ultimatum Game

We compute the set of symmetric Nash equilibria in the UG when $\alpha_1 = \alpha_2 = \alpha > 0$ and $\kappa_1 = \kappa_2 = \kappa > 0$. This extends [Alger and Weibull \(2012\)](#) that do so when $\alpha = 0$ and $\kappa = 0$. Let $x = (x_1, x_2)$ and $y = (y_1, y_2)$ denote the strategies of individuals 1 and 2, respectively. We normalize $v(0) \equiv 0$. Given that we focus on symmetric strategies $x_1 = y_1$ and $x_2 = y_2$, we distinguish when $x_1 > \frac{w}{2}$ or when $x_1 < \frac{w}{2}$. In the former, it is the *proposer* who suffers from behindness aversion, while in the latter it is the *responder*. When $x_1 \geq \frac{w}{2}$, the utility of individual 1's is given by:

$$\begin{aligned} u(x, y) &= \mathbf{1}_{\{x_1 \geq y_2\}} [(1 + \alpha - \kappa)v(w - x_1) - \alpha v(x_1)] \\ &+ \mathbf{1}_{\{y_1 \geq x_2\}} (1 - \kappa)v(y_1) \\ &+ \mathbf{1}_{\{x_1 \geq x_2\}} \kappa [v(w - x_1) + v(x_1)]. \end{aligned} \quad (17)$$

When $x_1 \leq \frac{w}{2}$, individual 1's utility is given by:

$$\begin{aligned} u(x, y) &= \mathbf{1}_{\{x_1 \geq y_2\}} (1 - \kappa)v(w - x_1) \\ &+ \mathbf{1}_{\{y_1 \geq x_2\}} [(1 + \alpha - \kappa)v(y_1) - \alpha v(w - y_1)] \\ &+ \mathbf{1}_{\{x_1 \geq x_2\}} \kappa [v(w - x_1) + v(x_1)]. \end{aligned} \quad (18)$$

[Alger and Weibull \(2012\)](#) characterize the Nash equilibrium set when $\alpha = 0$ and $\kappa > 0$:

Proposition 2. (in [Alger and Weibull, 2012](#)): *The set of homo-moralis strategies in this ultimatum-bargaining game, X_κ , is of the form:*

$$X_\kappa = \left\{ x \in [0, w]^2 : \text{either } x_2 \leq x_1 = \tau(\kappa) \text{ or } \tau(\kappa) < x_1 = x_2 \leq \rho(\kappa) \right\} \quad (19)$$

where $\tau(\kappa) \leq \frac{w}{2}$ is defined by $\tau(\kappa) = \min \{x \in [0, w] : v'(w - x) \geq \kappa v'(x)\}$ and $\rho(\kappa) \geq \frac{w}{2}$ is continuous and decreasing in κ .

³⁹The numerator is positive as $v(w - \underline{x}_2) - v(\underline{x}_2) > 0$, while the denominator is positive as $v' > 0$.

Therefore, there exist lower and upper bounds on the offers that can be sustained in equilibrium. When $x_1 > \tau(\kappa)$, any symmetric Nash equilibria must satisfy $x_1 = x_2$. On the other hand, when $x_1 = \tau(\kappa)$, any strategy $x = (x_1, x_2)$ with $x_1 = \tau(\kappa)$ and $x_2 \in [0, \tau(\kappa)]$ is a Nash equilibrium.⁴⁰

We extend this result by considering the case with $\alpha > 0$. We start by restricting attention to equilibria where offers and rejection thresholds are not higher than $\frac{w}{2}$ (i.e., $x_1 \leq \frac{w}{2}$). In this case, spite only enters individuals' maximization problem in the *responder* role (see 18). When incorporating $\alpha > 0$, individuals do not necessarily want to accept all positive offers, as some offers may give them a negative payoff. Let $\underline{x}_2(\kappa, \alpha) = \min\{x \in [0, w] : v(x)(1 + \alpha - \kappa) \geq \alpha v(w - x)\}$ (this is equivalent to the definition in the main text). Lemma 3 shows that there cannot exist a Nash equilibrium with $x_1^* < \underline{x}_2(\kappa, \alpha)$.

Lemma 3. *Let x^* be a symmetric Nash equilibrium. Then, $x_1^* \geq \underline{x}_2(\kappa, \alpha)$.*

Proof. By contradiction, suppose there exists x^* with $x_1^* < \underline{x}_2(\kappa, \alpha) < \frac{w}{2}$. We distinguish two cases depending on whether $x_1^* > x_2^*$ or if $x_1^* = x_2^* = x^*$. For the former, individual 1 is better off simply increasing his rejection threshold, as this does not affect the first and third term of equation (17), but increases the second one. For the latter, if $\kappa = 0$, then the individual sets $x_2^* = \underline{x}_2(\kappa, \alpha)$, given that he does not experience disutility when setting $x_1 < x_2$. When $\kappa > 0$, individual 1 can profitably deviate by selecting $x_1 = x_2 = x^* + \varepsilon$, where $\varepsilon > 0$. That is, the individual slightly increases their offer and rejection threshold. This improves individual 1's utility, as while the second term of equation (17) goes from negative to zero (as the individual 1 rejects individual 2's offer), the first and third term only change marginally. For the continuity of v , we can always find an $\varepsilon > 0$ sufficiently small such that the improvement of not accepting an offer below $\underline{x}_2(\kappa, \alpha)$ compensates for the marginal changes in the first and third terms. $\square$

We distinguish between two cases depending on whether (i) $\underline{x}_2(\kappa, \alpha) > \tau(\kappa)$ and (ii) $\underline{x}_2(\kappa, \alpha) \leq \tau(\kappa)$. When $\underline{x}_2(\kappa, \alpha) \leq \tau(\kappa)$, we have the same Nash equilibrium set characterized in [Alger and Weibull \(2012\)](#). The reason is that in such set the lowest offer possible, $\tau(\kappa)$, gives a positive material payoff to the responder. Hence, the individual does not have incentives to deviate from $(x_1^*, x_2^*) = (\tau(\kappa), \underline{x}_2(\kappa, \alpha))$.

On the other hand, when $\underline{x}_2(\kappa, \alpha) > \tau(\kappa)$, we have that the set of (symmetric) Nash

⁴⁰Therefore, the larger κ , the smaller the Nash equilibria set, similar to [Juan-Bartroli and Karagözoğlu \(2024\)](#).

equilibria is given by:

$$X_{\{\kappa, \alpha\}} = \{x \in [0, \frac{w}{2}]^2 : \underline{x}_2(\kappa, \alpha) \leq x_2 = x_1 \leq \frac{w}{2}\} \quad (20)$$

The reason is that when $\underline{x}_2(\kappa, \alpha) > \tau(\kappa)$, we cannot have a symmetric equilibrium with $x_2 < x_1$ (Claim 4 of [Alger and Weibull, 2012](#)). Intuitively, if individuals were to set a rejection threshold strictly below x_1 , they would be better off by decreasing their offer.

We now consider offers above $\frac{w}{2}$. In that case, let $\bar{x}_1(\kappa, \alpha) = \max\{x \in [0, w] : (1 - \kappa + \alpha)v(w - x) \geq v(x)\}$.

Lemma 4 shows that there cannot exist a (symmetric) Nash equilibrium with $x_1^* > \bar{x}_1(\kappa, \alpha)$.

Lemma 4. *Let x^* be a symmetric Nash equilibrium. Then, $x_1^* \leq \bar{x}_1(\kappa, \alpha)$.*

Proof. By contradiction, suppose there exists x^* with $x_1^* = x_2^* > \bar{x}_1(\kappa, \alpha)$. In that case, the first term in (17) is negative, while the second and third terms are positive. In that case, individual is better off by selecting, e.g., $x_1^* = x_2^* = \frac{w}{2}$. This increases the first term in (17) (it becomes zero instead of negative), does not affect the second term, and increases the third term. $\square$

Proposition 3 gives the set of Nash Equilibrium.

Proposition 3. *The set of (symmetric) Nash Equilibrium in the UG when individuals have both $\alpha > 0$ and $\kappa > 0$ is of the form:*

$$X_{\{\kappa, \alpha\}} = \{x \in [0, w]^2 : \max[\underline{x}_2(\kappa, \alpha), \tau(\kappa)] \leq x_2 = x_1 \leq \min[\bar{x}_1(\kappa, \alpha), \rho(\kappa)] \text{ and } x_2 \geq x_1 = \tau(\kappa) \text{ when } \tau(\kappa) > \underline{x}_2(\kappa, \alpha)\} \quad (21)$$

Proof. Follows from combining the previous lemmas. $\square$

Thus, the set of (symmetric) Nash equilibria shrinks when κ and α increase, and hence the lowest offers and rejection thresholds that can be sustained in equilibrium are increasing in κ and α . Figure 6 displays the set of (symmetric) Nash equilibria when (i) $\alpha = 0$ (in blue) and (ii) $\alpha > 0$ and $\underline{x}_2(\kappa, \alpha) > \tau(\kappa)$ (in red). When $\alpha = 0$, we have that the equilibrium can be either (i) $x_1^* = x_2^* \in [\tau(\kappa), \rho(\kappa)]$, or (ii) $x_1^* = \rho(\kappa)$ and $x_2^* \in [0, \rho(\kappa)]$. On the other hand, when $\alpha > 0$ and $\underline{x}_2(\kappa, \alpha) > \tau(\kappa)$, then we have that the equilibrium is $x_1^* = x_2^* \in [\underline{x}_2(\kappa, \alpha), \bar{x}_1(\kappa, \alpha)]$.

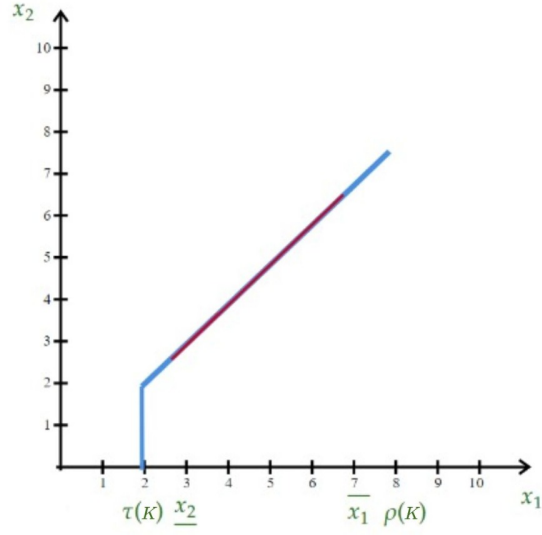


Figure 6: Set of (symmetric) Nash equilibria when (i) $\alpha = 0$ (in blue) and (ii) $\alpha > 0$ and $\underline{x}_2(\kappa, \alpha) > \tau(\kappa)$ (in red). $\tau(\kappa) = 2$, $\underline{x}_2(\kappa, \alpha) = 3$, $\bar{x}_1(\kappa, \alpha) = 7$, and $\rho(\kappa) = 8$.

Appendix C Simulations and Figures

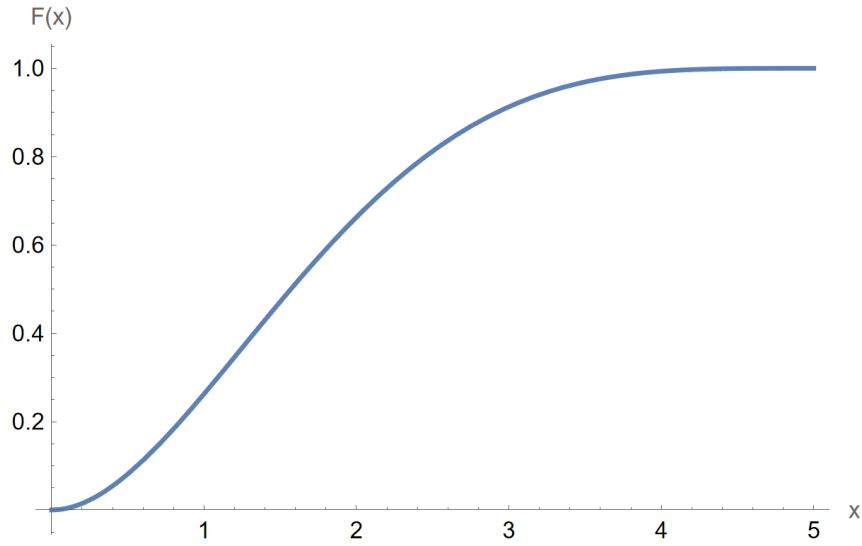


Figure 7: Distribution offers and rejection thresholds $F(x) = F_{[0,w/2]}(2,4)$.

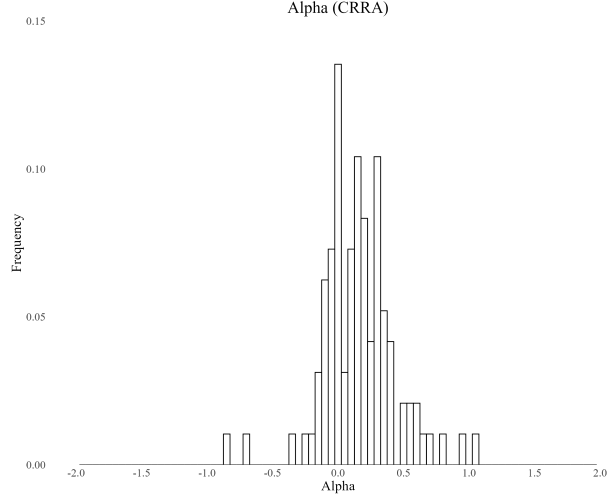


Figure 8: Distribution of individual estimates of α_i in [van Leeuwen and Alger \(2024\)](#), CRRA estimates, for $\alpha, \beta \in [-2, 2]$ and $\kappa \in [0, 1]$.

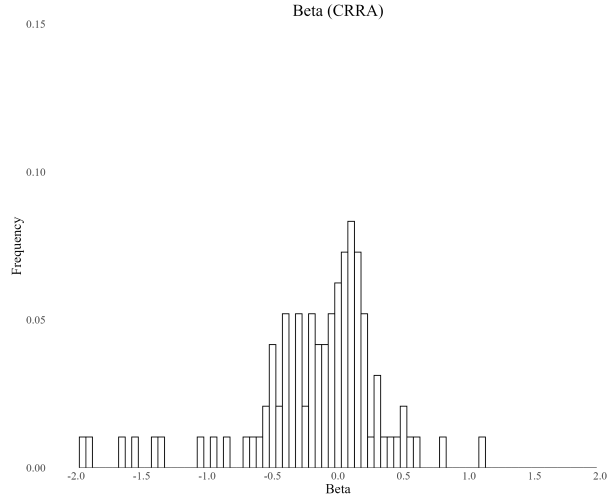


Figure 9: Distribution of individual estimates of β_i in [van Leeuwen and Alger \(2024\)](#), CRRA estimates, for $\alpha, \beta \in [-2, 2]$ and $\kappa \in [0, 1]$.

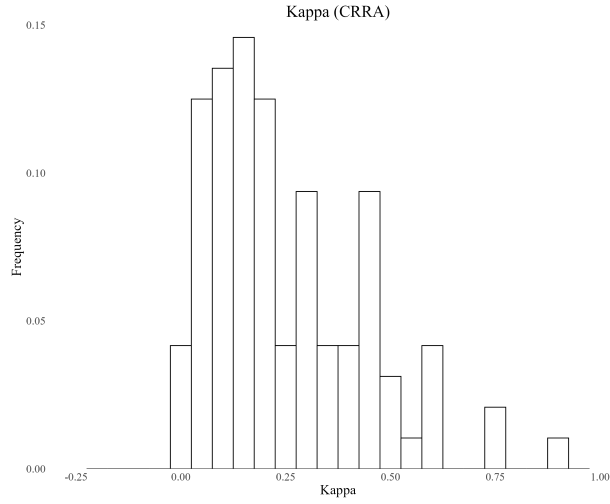


Figure 10: Distribution of individual estimates of κ_i in [van Leeuwen and Alger \(2024\)](#), CRRA estimates, for $\alpha, \beta \in [-2, 2]$ and $\kappa \in [0, 1]$.

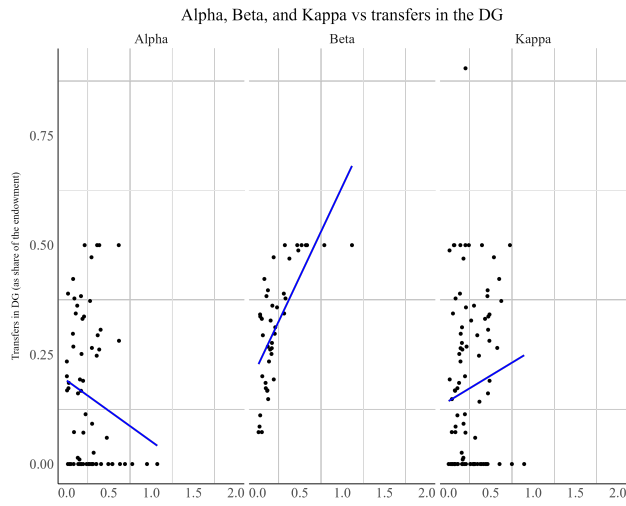


Figure 11: Transfers in the DG and parameter estimates. [van Leeuwen and Alger \(2024\)](#) CRRA estimates, for $\alpha, \beta \in [-2, 2]$ and $\kappa \in [0, 1]$.

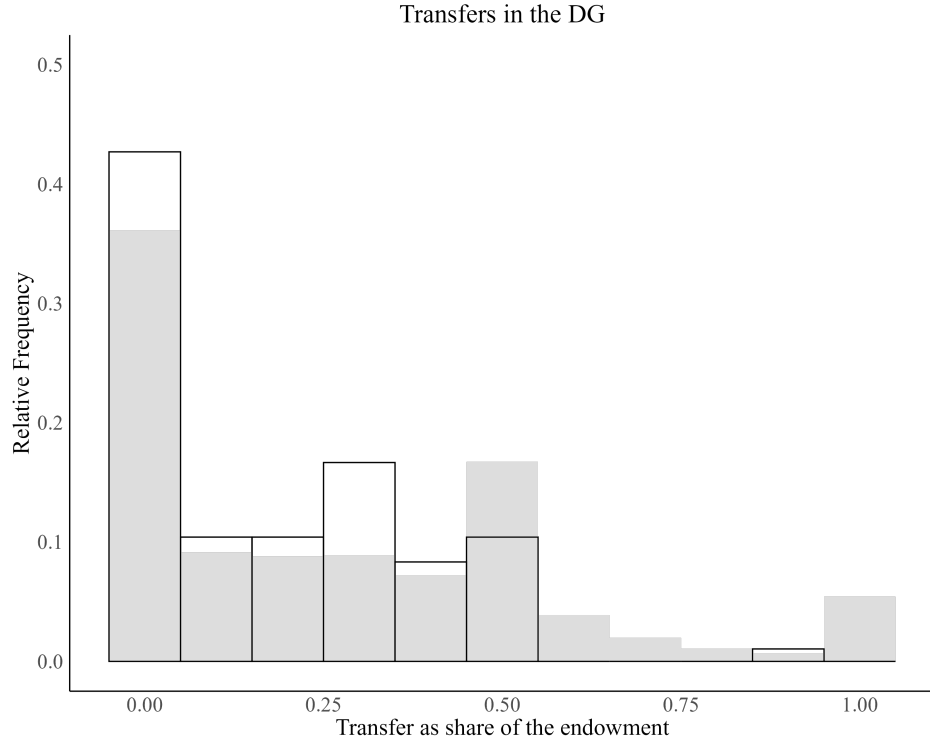


Figure 12: DG transfers reported in [Engel \(2011\)](#) –in gray– and transfers based on [van Leeuwen and Alger \(2024\)](#) CRRA estimates, for $\alpha, \beta \in [-2, 2]$ and $\kappa \in [0, 1]$.

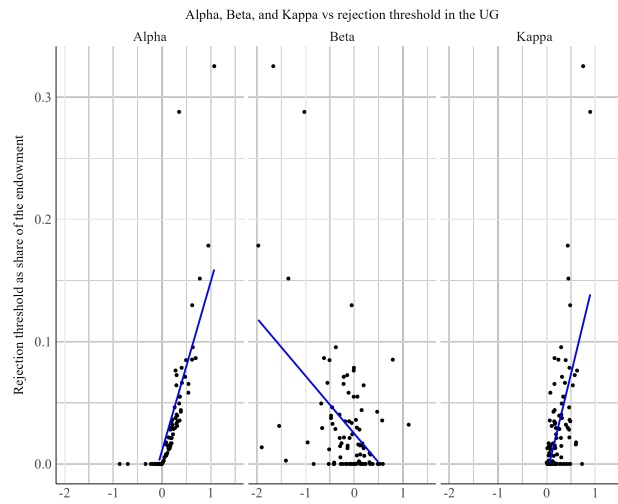


Figure 13: Rejection thresholds and parameter estimates. [van Leeuwen and Alger \(2024\)](#) CRRA estimates, for $\alpha, \beta \in [-2, 2]$ and $\kappa \in [0, 1]$.

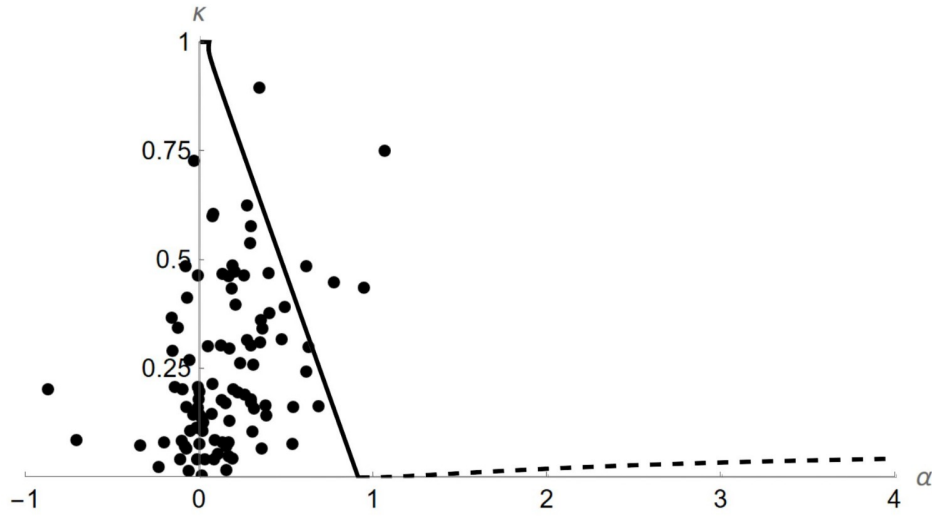


Figure 14: Optimal strategy in Figure 1 with individual estimates.

Variable	Min	Q1	Median	Q3	Max	Mean	Obs.
Donation DG	0.00 (0.00)	0.00 (0.00)	6.62 (0.11)	18.60 (0.32)	53.20 (0.90)	10.20 (0.17)	96
Threshold UG	0.00 (0.00)	0.00 (0.00)	0.87 (0.01)	2.34 (0.04)	19.10 (0.33)	1.90 (0.03)	96

Table 3: Descriptive Statistics of transfers in DG and Rejection Threshold in UG. [van Leeuwen and Alger \(2024\)](#) core sample, CRRA estimates, for $\alpha, \beta \in [-2, 2]$ and $\kappa \in [0, 1]$. Share of the endowment in parentheses.

Variable	Min	Q1	Median	Q3	Max	Mean	Obs.
Donation DG	0.00 (0.00)	0.00 (0.00)	5.17 (0.09)	19.90 (0.34)	58.80 (1.00)	10.40 (0.18)	96
Threshold UG	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.41 (0.01)	2.62 (0.04)	0.32 (0.01)	96

Table 4: Descriptive Statistics of transfers in DG and Rejection Threshold in UG. [van Leeuwen and Alger \(2024\)](#) core sample, CRRA estimates with only α and β , for $\alpha, \beta \in [-2, 2]$. Share of the endowment in parentheses.

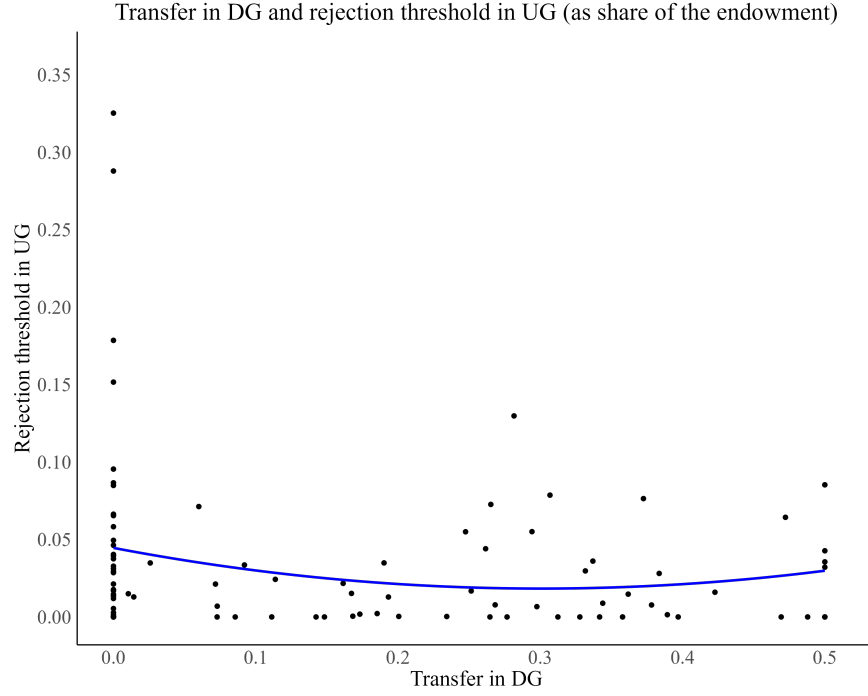


Figure 15: Transfer in DG and constrained optimal rejection threshold in UG. Both expressed as share of the endowment. [van Leeuwen and Alger \(2024\)](#) core sample, CRRA estimates, for $\alpha, \beta \in [-2, 2]$ and $\kappa \in [0, 1]$. $N = 96$.

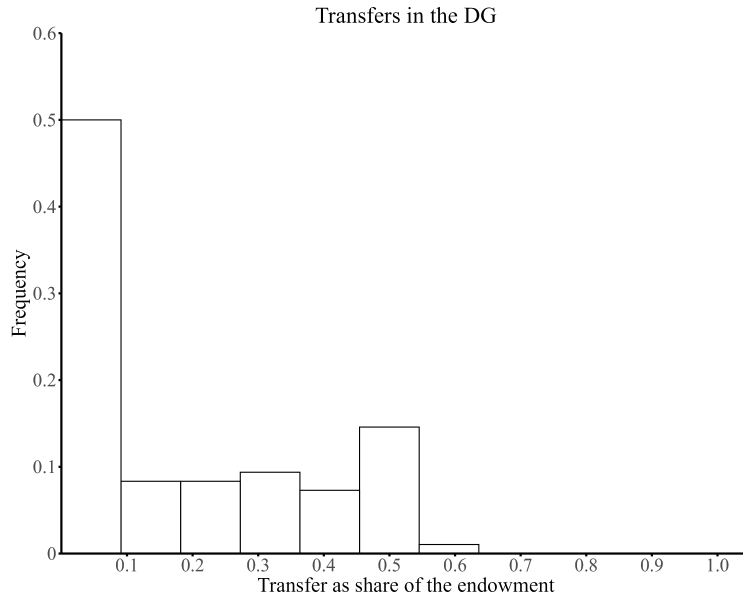


Figure 16: Predicted share transferred in the Dictator Game. [van Leeuwen and Alger \(2024\)](#) CRRA estimates with only α and β ; $\alpha, \beta \in [-2, 2]$ and $w = 58.8$ (10 euros). $N = 96$.

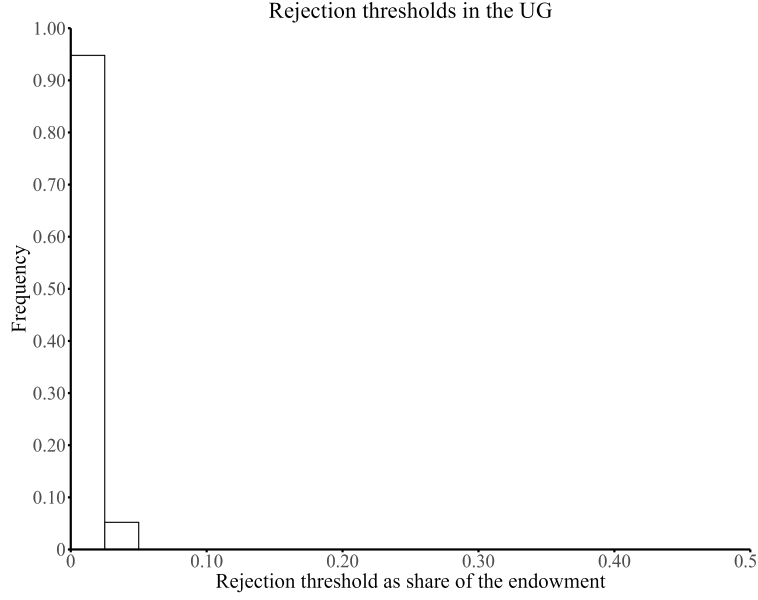


Figure 17: Constrained optimal rejection threshold $\underline{x}_2$ in the UG. [van Leeuwen and Alger \(2024\)](#) CRRA estimates with only α and β ; $\alpha, \beta \in [-2, 2]$ and $w = 58.8$ (10 euros). $N = 96$.

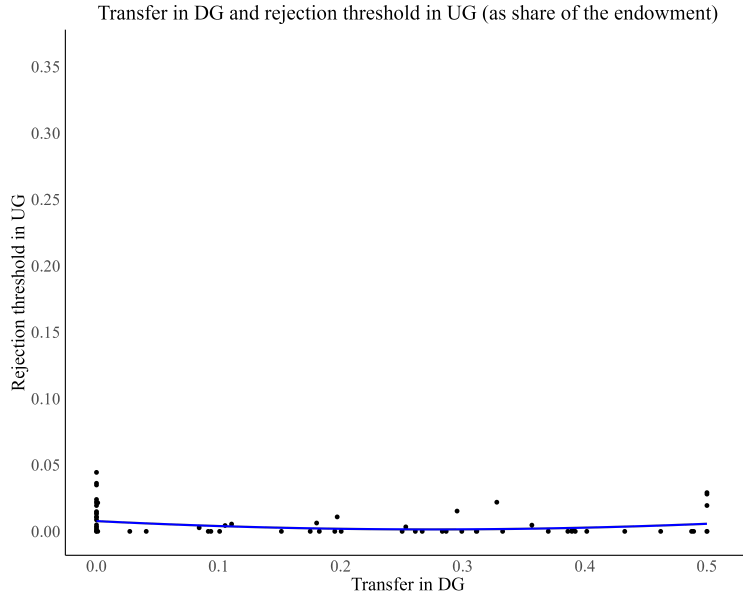


Figure 18: Transfer in DG and constrained optimal rejection threshold in UG. [van Leeuwen and Alger \(2024\)](#) CRRA estimates with only α and β ; $\alpha, \beta \in [-2, 2]$ and $w = 58.8$ (10 euros). $N = 96$.

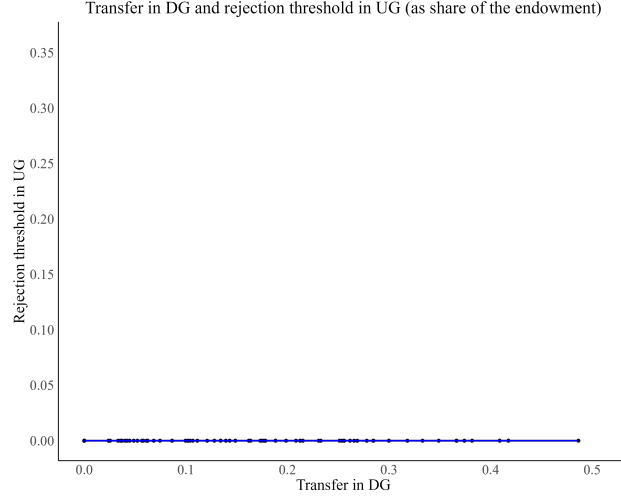


Figure 19: Transfer in DG and constrained optimal rejection threshold in UG. [van Leeuwen and Alger \(2024\)](#) CRRA estimates with only $\kappa \in [0,1]$, when $w = 58.8$ (10 euros).

Appendix D Comparison with [Brañas-Garza et al. \(2014\)](#)

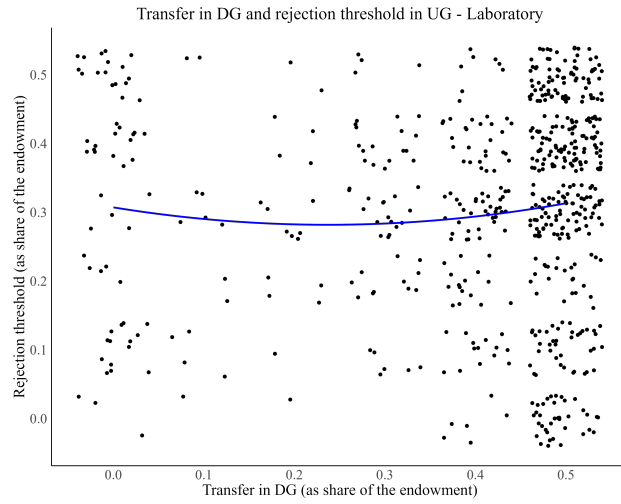


Figure 20: Transfer in DG and rejection threshold in UG in “Laboratory” study reported in [Brañas-Garza et al. \(2014\)](#)

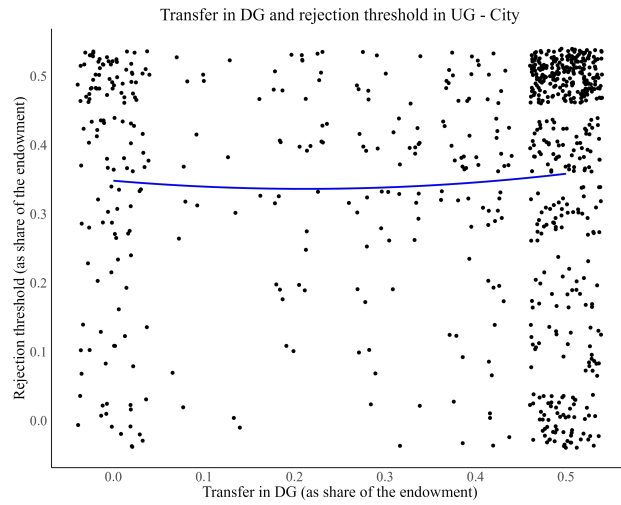


Figure 21: Transfer in DG and rejection threshold in UG in “City” study reported in [Brañas-Garza et al. \(2014\)](#)

References

- Aina, C., P. Battigalli, and A. Gamba (2020). Frustration and anger in the ultimatum game: An experiment. *Games and Economic Behavior* 122, 150–167.
- Alger, I. (2023). Evolutionarily stable preferences. *Philosophical Transactions of the Royal Society B* 378(1876), 20210505.
- Alger, I. and J.-F. Laslier (2022). Homo moralis goes to the voting booth: Coordination and information aggregation. *Journal of Theoretical Politics* 34(2), 280–312. SAGE Publications Sage UK: London, England.
- Alger, I. and J. I. Rivero-Wildemaue (2024). Doing the right thing (or not) in a lemons-like situation: on the role of social preferences and kantian moral concerns. *arXiv preprint arXiv:2405.13186*.
- Alger, I. and J. W. Weibull (2012). Homo moralis—preference evolution under incomplete information and assortative matching. TSE Working Paper 12-281, Toulouse School of Economics.
- Alger, I. and J. W. Weibull (2013). Homo moralis—preference evolution under incomplete information and assortative matching. *Econometrica* 81(6), 2269–2302.
- Alger, I. and J. W. Weibull (2016). Evolution and kantian morality. *Games and Economic Behavior* 98, 56–67.
- Alger, I. and J. W. Weibull (2017). Strategic behavior of moralists and altruists. *Games* 8(3), 38.
- Alger, I., J. W. Weibull, and L. Lehmann (2020). Evolution of preferences in structured populations: genes, guns, and culture. *Journal of Economic Theory* 185, 104951.
- Andreoni, J. and B. D. Bernheim (2009). Social image and the 50–50 norm: A theoretical and experimental analysis of audience effects. *Econometrica* 77(5), 1607–1636.
- Aquino, K. and A. Reed (2002). The self-importance of moral identity. *Journal of Personality and Social Psychology* 83(6), 1423–1440.
- Becker, G. S. (1976). Altruism, egoism, and genetic fitness: Economics and sociobiology. *Journal of Economic Literature* 14(3), 817–826. JSTOR.

- Bellemare, C., S. Kröger, and A. Van Soest (2008). Measuring inequity aversion in a heterogeneous population using experimental decisions and subjective probabilities. *Econometrica* 76(4), 815–839.
- Biernacki, C., G. Celeux, and G. Govaert (2000). Assessing a mixture model for clustering with the integrated completed likelihood. *IEEE transactions on pattern analysis and machine intelligence* 22(7), 719–725.
- Binmore, K. G. (1994). *Game theory and the social contract: just playing*, Volume 2. MIT press.
- Brañas-Garza, P., A. M. Espín, F. Exadaktylos, and B. Herrmann (2014). Fair and unfair punishers coexist in the ultimatum game. *Scientific reports* 4(1), 1–4.
- Brekke, K. A., S. Kverndokk, and K. Nyborg (2003). An economic model of moral motivation. *Journal of Public Economics* 87(9-10), 1967–1983.
- Bruhin, A., E. Fehr, and D. Schunk (2019). The many faces of human sociality: Uncovering the distribution and stability of social preferences. *Journal of the European Economic Association* 17(4), 1025–1069.
- Butler, J. V., P. Giuliano, and L. Guiso (2015). Trust, values, and false consensus. *International Economic Review* 56(3), 889–915.
- Camerer, C. F. (2003). Behavioral game theory: Plausible formal models that predict accurately. *Behavioral and Brain Sciences* 26(2), 157–158.
- Candelo, N., C. Eckel, and C. Johnson (2019). The proposer’s behavior in the ultimatum game in 11 mexican villages. *Economics Letters* 177, 5–8.
- Capraro, V. and D. G. Rand (2018). Do the right thing: Experimental evidence that preferences for moral behavior, rather than equity or efficiency per se, drive human prosociality. *Judgment and Decision Making* 13(1), 99–111.
- Capraro, V. and I. Rodriguez-Lara (2025). Moral preferences in ultimatum and impunity games. *Journal of Behavioral and Experimental Economics*, 102371.
- Çelen, B., A. Schotter, and M. Blanco (2017). On blame and reciprocity: Theory and experiments. *Journal of Economic Theory* 169, 62–92.
- Celeux, G. and G. Soromenho (1996). An entropy criterion for assessing the number of clusters in a mixture model. *Journal of classification* 13, 195–212.

- Chen, D. L. and M. Schonger (2022). Social preferences or sacred values? Theory and evidence of deontological motivations. *Science Advances* 8(19). American Association for the Advancement of Science.
- Cochard, F., J. Le Gallo, N. Georgantzis, and J.-C. Tisserand (2021). Social preferences across different populations: Meta-analyses on the ultimatum game and dictator game. *Journal of Behavioral and Experimental Economics* 90, 101613.
- Daley, B. and P. Sadowski (2017). Magical thinking: A representation result. *Theoretical Economics* 12(2), 909–956.
- Dawes, R. M. (1989). Statistical criteria for establishing a truly false consensus effect. *Journal of Experimental Social Psychology* 25(1), 1–17.
- Dizarlar, A. and E. Karagözoğlu (2023). Kantian equilibria of a class of nash bargaining games. *Journal of Public Economic Theory* 25(4), 867–891.
- Drouvelis, M. and E. M. Gavassa-Pérez (2024). Redistribution Over Gains and Losses: Social Preferences and Moral Rules. Technical report. Working Paper.
- Dufwenberg, M. and G. Kirchsteiger (2004). A theory of sequential reciprocity. *Games and Economic Behavior* 47(2), 268–298.
- Eichner, T. and R. Pethig (2021). Climate policy and moral consumers. *The Scandinavian Journal of Economics* 123(4), 1190–1226. Wiley Online Library.
- Engel, C. (2011). Dictator games: A meta study. *Experimental economics* 14(4), 583–610.
- Erev, I. and A. E. Roth (1998). Predicting how people play games: Reinforcement learning in experimental games with unique, mixed strategy equilibria. *American Economic Review*, 848–881.
- Falk, A., E. Fehr, and U. Fischbacher (2005). Driving forces behind informal sanctions. *Econometrica* 73(6), 2017–2030.
- Fehr, E. and U. Fischbacher (2003). The nature of human altruism. *Nature* 425(6960), 785–791.
- Fehr, E. and K. M. Schmidt (1999). A theory of fairness, competition, and cooperation. *The Quarterly Journal of Economics* 114(3), 817–868.
- Gavassa-Pérez, E. M. (2022). Moral rules and social preferences in cooperation problems. *Working Paper*.

- Gintis, H., S. Bowles, R. Boyd, and E. Fehr (2003). Explaining altruistic behavior in humans. *Evolution and Human Behavior* 24(3), 153–172.
- Güth, W. and M. G. Kocher (2014). More than thirty years of ultimatum bargaining experiments: Motives, variations, and a survey of the recent literature. *Journal of Economic Behavior & Organization* 108, 396–409.
- Güth, W., R. Schmittberger, and B. Schwarze (1982). An experimental analysis of ultimatum bargaining. *Journal of Economic Behavior & Organization* 3(4), 367–388.
- Harsanyi, J. C. (1955). Cardinal welfare, individualistic ethics, and interpersonal comparisons of utility. *Journal of Political Economy* 63(4), 309–321.
- Herrmann, B., C. Thoni, and S. Gächter (2008). Antisocial punishment across societies. *Science* 319(5868), 1362–1367.
- Iriberri, N. and P. Rey-Biel (2011). The role of role uncertainty in modified dictator games. *Experimental Economics* 14, 160–180.
- Iriberri, N. and P. Rey-Biel (2013). Elicited beliefs and social information in modified dictator games: What do dictators believe other dictators do? *Quantitative Economics* 4(3), 515–547.
- Juan-Bartroli, P. (2023). On injunctive norms: Theory and experiment. *Working Paper*.
- Juan-Bartroli, P. and E. Karagözoğlu (2024). Moral preferences in bargaining. *Economic Theory*, 1–24. Springer.
- Kant, I. (1785). *Grundlegung zur Metaphysik der Sitten*. Riga: Johann Friedrich Hartknoch.
- Karagözoğlu, E. and K. Keskin (2018). Time-varying fairness concerns, delay, and disagreement in bargaining. *Journal of Economic Behavior & Organization* 147, 115–128.
- Kimbrough, E. O. and A. Vostroknutov (2016). Norms make preferences social. *Journal of the European Economic Association* 14(3), 608–638.
- Levine, D. K. (1998). Modeling altruism and spitefulness in experiments. *Review of Economic Dynamics* 1(3), 593–622.
- McLachlan, G. J., S. X. Lee, and S. I. Rathnayake (2019). Finite mixture models. *Annual review of statistics and its application* 6(1), 355–378.

- Miettinen, T., M. Kosfeld, E. Fehr, and J. Weibull (2020). Revealed preferences in a sequential prisoners' dilemma: A horse-race between six utility functions. *Journal of Economic Behavior & Organization* 173, 1–25. Elsevier.
- Mui, V.-L. (1995). The economics of envy. *Journal of Economic Behavior & Organization* 26(3), 311–336.
- Muñoz Sobrado, E. (2022). Taxing Moral Agents. *CESifo Working Paper*.
- Rabin, M. (1993). Incorporating fairness into game theory and economics. *The American Economic Review*, 1281–1302.
- Rawls, J. (1971). *A Theory of Justice*. Cambridge, MA: Harvard University Press.
- Rivero-Wildemauwe, J. (2023). Moral motivations in sequential buyer-seller interactions with adverse selection. Technical report, THEMA (THéorie Economique, Modélisation et Applications), CY Cergy Paris Université.
- Rivero-Wildemauwe, J. I. (2025). Trade among moral agents with information asymmetries. Technical Report arXiv:2505.20551, arXiv. Available at <https://arxiv.org/abs/2505.20551>.
- Roemer, J. E. (2010). Kantian Equilibrium. *Scandinavian Journal of Economics* 112(1), 1–24.
- Roemer, J. E. (2015). Kantian Optimization: A microfoundation for cooperation. *Journal of Public Economics* 127, 45–57.
- Ross, L., D. Greene, and P. House (1977). The “false consensus effect”: An egocentric bias in social perception and attribution processes. *Journal of Experimental Social Psychology* 13(3), 279–301.
- Roth, A. E. (2007). Repugnance as a Constraint on Markets. *Journal of Economic Perspectives* 21(3), 37–58. American Economic Association.
- Salonia, E.-M. (2023). A Foundation for Universalization in Games. *Working Paper*.
- Sarkisian, R. (2017). Team Incentives under Moral and Altruistic Preferences: Which Team to Choose? *Games* 8(3), 37. Multidisciplinary Digital Publishing Institute.
- Sen, A. K. (1977). Rational fools: A critique of the behavioral foundations of economic theory. *Philosophy & public affairs*, 317–344.

- Shafir, E. and A. Tversky (1992). Thinking through uncertainty: Nonconsequential reasoning and choice. *Cognitive Psychology* 24(4), 449–474.
- Smith, V. L. and B. J. Wilson (2019). *Humanomics: Moral sentiments and the wealth of nations for the twenty-first century*. Cambridge University Press.
- van Leeuwen, B. and I. Alger (2024). Estimating Social Preferences and Kantian Morality in Strategic Interactions. *Journal of Political Economy Microeconomics*.
- Xiao, E. and D. Houser (2005). Emotion expression in human punishment behavior. *Proceedings of the National Academy of Sciences* 102(20), 7398–7401.